Allocation Before the Clock: Graph, Set and Game Theory of Professional League of Legends Objective Setups

Zheng Gao[†], **Zhenyu Liu, Yifan Yang, Xiaoyu Li**
UNSW Sydney [†]Corresponding author: zheng.gao1@unsw.edu.au

Abstract

Before every neutral objective in professional *League of Legends*, each team decides which of its five players walk to the pit, which hold a lane, and when to commit. The leading analysis of such setups, TAGNet for *Counter-Strike 2*, clusters graph-network embeddings of player positions and reads cluster win rates as tactical quality. We treat a setup as an allocation on a walled map: formations are set partitions on the lattice Π_5, control is a geodesic Voronoi split of a navigation graph, and commitment is an allocation game solved as a quantal response equilibrium priced in win probability. On all 164 games of MSI and Worlds 2025 (2,994 team-setups, minimap tracks validated against Riot positions), trailing teams leave the pit empty far more often than leading teams, by +34.8 pp at the first baron and +30.4 pp at Atakhan. Our model, HASSE-QRE, finds six setup patterns that whole-model series-bootstrap refits reproduce (ARI 0.819, comparable to 0.813 for TAGNet's K-Means step alone) and indicates that teams price fights, travel and committed players, not the objective itself (weight 0.018; fixing it to one costs 59.9 log-likelihood points). The equilibrium reproduces a quarter of the commitment gap, almost all of it through state variables rather than its strategic channels. Weighting a team-level contrast by exposure moves it across $p = 0.05$ (0.046 against 0.052). We release the tracks, setups and code.

Keywords: esports analytics, League of Legends, set partitions, quantal response equilibrium, Lanchester laws, Colonel Blotto, spatiotemporal tracking

The question. *When a neutral objective is about to spawn on a clock that both teams can read, how does each team allocate its five players on a walled map, and what do teams actually price when they decide how many to commit?*

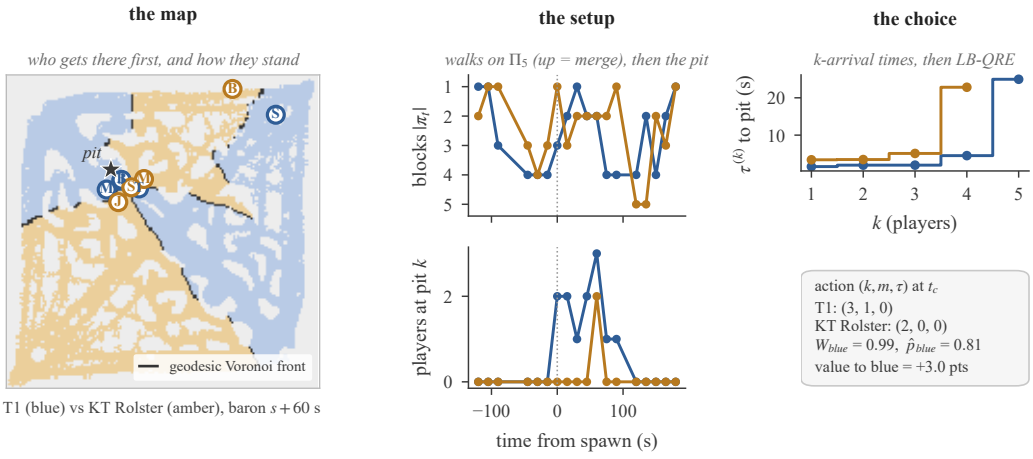


Figure 1: **The paper in one picture.** One real setup (game 4 of the Worlds 2025 final, first baron, 60 s after the spawn). *The map:* walkable graph (white: walls) coloured by which team reaches each cell first, with each formation drawn as the blocks of a set partition. *The setup:* each team's walk on the partition lattice Π_5 (up = fewer blocks) and its players at the pit. *The choice:* k-arrival times $\tau^{(k)}$ and the LB-QRE action (k, m, τ) at the commitment time, with win probability W, fight probability $\hat{p}$ and the objective's value.

1 Introduction

Neutral objectives in *League of Legends* (LoL) appear on a public clock. The first dragon spawns at 5:00, the void grubs at 8:00, the Rift Herald at 15:00, Atakhan at 20:00 and Baron Nashor at 25:00, and both teams know these times to the second. In the minute before each spawn the two teams move their ten players into position: some walk to the pit, some keep pushing a lane, some wait in the jungle in between. This setup decides who fights for the objective, with how many players, and whether the other side trades it for towers elsewhere. Professional coaches plan it in detail. Esports analytics has measured almost none of it.

The closest work is TAGNet (Zhang et al., 2026), which studies how defenders set up after the freeze time in professional *Counter-Strike 2*. TAGNet embeds the five defenders as a complete graph with a graph attention network (Veličković et al., 2018) and an untrained recurrent layer, clusters the 64-dimensional embeddings with K-Means at a fixed $K = 6$, and reads the win rate of each cluster as the quality of a tactic. Three steps of that pipeline make the conclusions fragile. The graph carries no map: straight-line distances ignore walls. The context variables (economy, score) are fed into the representation and later 'discovered' to be associated with it. And comparisons of trailing and leading teams average over teams rather than over rounds.

We take a different view. A setup is an *allocation on a walled map*. Distances are geodesic on a navigation graph built from Riot's own position data, so a formation is the set of connected components of a proximity graph, that is, a set partition of the five roles, and the formations form the partition lattice Π_5 with 52 elements. Map control is the geodesic Voronoi split of that graph, and readiness is a vector of k-arrival times. The decision is how many players to commit to the pit, how many to a second front, and whether to be there before the spawn. Fight outcomes follow a Lanchester-type law (Lanchester, 1916) written as a Tullock contest (Tullock, 1980), the second front is a Blotto-type contest (Roberson, 2006), and every outcome is priced in win probability by a surface W fitted on game results only. Figure 1 shows all of these objects on one real baron setup from the Worlds 2025 final.

The first finding is about behaviour. Across all 164 games of MSI and Worlds 2025, trailing teams leave the pit empty far more often than leading teams once the objective is up. At the first baron the gap reaches +34.8 pp [18.7, 50.0] at s+45 s, at Atakhan +30.4 pp [15.0, 46.5], and at respawned dragons +33.7 pp [24.9, 41.5] (Figures 5 and 6). Every headline contrast keeps its sign at three pit radii and survives false-discovery control. The same contrast on Riot's exact 60-second frames gives +19.7 pp [11.4, 29.9] at the first baron. The dragon gradient of our kill-anchored pilot (13% of trailing vs 43% of leading non-takers contested) disappears once spawns rather than kills anchor the window: it was an artefact of selecting on the outcome.

The second finding is about structure. HASSE, a sticky hidden Markov model whose emissions live on Π_5, describes each setup as a walk on the Hasse diagram with a pattern-specific centre path, rigidity and inertia. Held-out likelihood improves up to $K = 8$ with steps that shrink to 0.37 nats per setup, and six patterns carry the structure (Figure 8). Two of them, used almost only at early river objectives, stay spread out; two gather four players around the pit at the spawn and dominate the late game. Whole-model series-bootstrap refits of HASSE agree at ARI 0.819 (27 series-bootstrap resamples), comparable to re-running only TAGNet's K-Means step on fixed embeddings (0.813); five independent HASSE fits, each the best of ten restarts, agree at 0.890, whereas single TAGNet runs on the same V-layer tracks agree at 0.421 across seeds and at 0.086 with TAGNet's own counting baseline (Figure 9).

The third finding is about incentives. LB-QRE, a heterogeneous logit quantal response equilibrium (McKelvey and Palfrey, 1995) of the allocation game estimated by nested pseudo-likelihood (Aguirregabiria and Mira, 2007), finds that teams respond to fight values, travel and the cost of each committed player, and barely to the objective's own value: its weight is $\nu = 0.018$ [0.013, 0.026], and fixing $\nu = 1$ (pure win-probability maximisation) costs 59.9 log-likelihood points (likelihood ratio about 120; bootstrap mean 62.9 [40.1, 92.8]). The equilibrium reproduces about a quarter of the observed trailing-minus-leading commitment gap (−4.7 vs −19.2 pp), but almost all of that comes from state

Table 1: **The core set covers every game of two 2025 events.** The core video set. Every game of MSI 2025 and Worlds 2025 under one set of objective rules.

Event	Games	Series	Teams	Spawns
MSI 2025	80	19	10	740
Worlds 2025 Main Event	80	40	16	740
Worlds 2025 Play-In	4	1	2	36
total	164	60	19	1,516

variables (alive counts, the gold term, travel and feasibility): with every strategic channel off the gap is −5.1 pp, so the four channels add +0.5 pp net. Within the model, the curvature of W pushes trailing teams towards contesting (+1.50 pp) and team behaviour pulls them away (−1.51 pp), while the Lanchester and second-front channels are small (Figure 11). Fights punish the smaller side hard: the larger side loses the next player far less often than the square law predicts (0.037 vs 0.375 at 5v3).

These results rest on a measurement layer that we validate end to end: a broadcast-clock reader aligns every video to game time, the minimap is registered to the navigation graph, and a per-game tracker is scored against Riot positions at held-out minutes (median error 177 map units on a map 14,820 units wide). In summary, we contribute (i) a theory-defined measurement of objective setups (navigation graph, geodesic Voronoi control, k-arrival order statistics, partitions on Π_5) with validated extraction from broadcast video; (ii) HASSE-QRE, a two-layer model whose discovery layer lives on the partition lattice and whose explanation layer is an estimable allocation game in win-probability units; (iii) the findings above, including the negative ones, with an estimator audit that shows how the choice of estimator can flip significance; and (iv) an open dataset of 1,516 spawns, 2,994 team-setups and 2-fps tracks, with an appendix that turns narratives about T1 and Faker into measured quantities.

2 Data and the Spawn Clock

Games. The core set is every game of the 2025 Mid-Season Invitational and the 2025 World Championship, 164 games in 60 series between 19 teams (Table 1). Both events were played on patches 25.13 and 25.20 under the same objective rules, so one spawn clock covers all games. For each game we hold three sources: the broadcast video, Riot's match-v5 timeline (Riot Games) with exact positions of all ten players every 60 s and millisecond-stamped kills and objectives, and Leaguepedia (Leaguepedia) metadata (roles, draft, series). An archive of 2,678 professional games with v5 timelines (2023–2026) is used only to fit outcome models: the win surface, event values and the navigation graph. It never sees a setup choice.

The spawn clock. Table 2 lists the rules. We infer every spawn time from the rules and the kill events and check that no kill precedes its inferred spawn; none does across the 1,516 spawns. Anchoring at the spawn rather than at the kill is the central design choice of this paper. The spawn time is fixed by rule and known to both teams, so a window anchored there does not depend on what the teams did; a window anchored at the kill selects on who took the objective and when. After the spawn we keep a team-snapshot only while the objective is still alive (a risk set), because a snapshot taken after the kill describes the aftermath rather than the setup.

Units. A *team-setup* is one team at one spawn, 2,994 in total. We observe it on a grid of snapshots every 15 s from $s-120$ to $s+180$ s, where s is the spawn time. Two layers share these units. The **V layer** uses 2-fps minimap tracks from the broadcast (Section 3) and covers 1,497 of 1,516 spawns (the rest lack a live track), 2,994 team-setups, on a grid of 21 snapshots of which a median 90% are observed; missing snapshots are treated as missing. The **R layer** uses only Riot's exact 60-s frames and covers the fixed-time first spawns (dragon, grubs, herald, Atakhan, baron) at $s-60$, s and $s+60$. The R layer has no measurement error and coarse time; the V layer has fine time and validated measurement error. Findings that matter are checked on both.

Table 2: **Every objective has a fixed spawn clock, and no kill precedes its spawn.** The spawn clock (patch 25.09–25.20 rules) and the resulting units. Delay is the median spawn-to-kill time; the last column gives the share of tracked positions that were detected, interpolated or pinned to a V5 minute.

Objective	First spawn	Respawn	Spawns	Taken	Delay (s)	Setups	det/int/pin
Elemental dragon	5:00	kill + 5:00 until a soul	773	743	49	1,526	0.75/0.20/0.06
Elder dragon	soul + 6:00	kill + 6:00	27	17	44	52	0.67/0.30/0.00
Void grubs	8:00	one camp of 3, no respawn	164	164	21	322	0.71/0.10/0.19
Rift Herald	15:00	once	164	164	50	328	0.72/0.16/0.12
Atakhan	20:00	once (side set by play)	164	164	109	328	0.71/0.21/0.08
Baron Nashor	25:00	kill + 6:00	224	181	117	438	0.69/0.27/0.03

Gold state. A team is *trailing*, *even* or *leading* by its gold difference at $s-120$ s divided by the archive standard deviation of the gold difference at that game minute, with cut points ± 0.5. The state is fixed before the setup begins, so it cannot respond to the setup. It is measured at $s-120$ s on the V layer, at $s-60$ s on the R layer (the first exact frame) and, for LB-QRE, at the last Riot frame at least 30 s before the spawn.

Objective classes. Mixture weights in the discovery layer are stratified by three classes that differ in timing and map: early river objectives (dragons, grubs, herald), Atakhan, and late objectives (baron, elder). In the descriptive analysis respawned dragons and barons form their own groups because their spawn time depends on play.

3 The Measurement Layer

The measurement layer turns video into positions and positions into the objects the models use. It has no learned parameters beyond the per-game detectors, and every step is validated against Riot data.

3.1 From broadcast to game coordinates

Clock alignment. We crop the broadcast's game clock at 10 fps and read it with a digit classifier trained on one game, labelled by a single hand-checked anchor reading and the clock's own progression. Each clock tick yields one pair (video time, game time); a robust fit over the roughly one thousand ticks of a game gives the offset of each live segment and separates live play from replays, which show earlier times for a few seconds and then return. All 166 videos (the 164 core games and two control videos) align (Figure 2a); the median share of frames carrying a live game time is 79.2%, the median misread rate 0.06%, and the last live reading falls a median -4.2 s before the official game end, the length of the closing camera sequence. 23 replay segments are detected and excluded.

Minimap registration. The minimap is an orthographic projection, so four numbers (scale and offset per axis) map pixels to map coordinates. The per-pixel median of a game's minimap frames removes the moving icons and leaves the static map; we register it to the walkable mask of the navigation graph by maximising their correlation. Two independent crops of the same game agree to within 0.04 pixels, and Riot's tower coordinates and per-minute player positions land on their icons.

Tracking. Each game has ten known champions, and the v5 timeline gives every champion's exact position once a minute. We therefore train detectors per game: for each champion a linear discriminant $w = \Sigma^{-1}(\mu_{\text{champ}} - \mu_{\text{bg}})$, where Σ is the covariance of random minimap patches, so static structures such as towers fall into the background distribution without labels. Scores are calibrated to likelihood ratios. A Viterbi pass per champion chooses among detection peaks, a 'held' state (icon occluded, position kept) and a 'dead' state that is allowed only inside death windows from the kill events; transitions are priced by geodesic distance against the maximum movement speed.

Validation. We validate on a harder task than the production run: detectors are trained and tracks pinned only at even minutes, and the tracker is scored at odd minutes, a full 60 s from the nearest pin

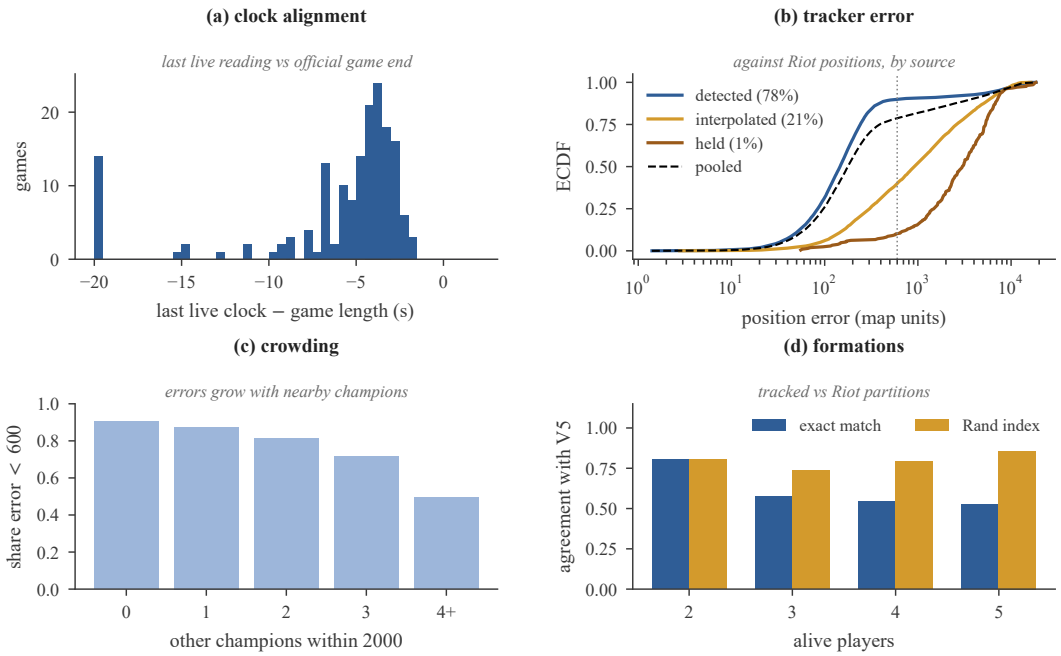


Figure 2: **The measurement layer is validated end to end.** (a) Last live clock reading minus official game length for all aligned videos. (b) Tracker position error against Riot v5 at held-out odd minutes, by position source; the dotted line marks 600 units. (c) Share of errors below 600 units by local crowding (other champions within 2,000 units). (d) Agreement of tracked formations with v5 formations: exact partition match and Rand index, by number of alive players.

(production pins every minute, at most 30 s away). Over 23,286 champion-minutes the median error is 177 units and 0.785 of errors are below 600 units (Table 5, Figure 2). Detected positions (78% of all) have median error 146 and 0.900 below 600; interpolated positions (21%) have median 921. Error grows with crowding: the share below 600 falls from 0.905 with no other champion within 2,000 units to 0.493 with four or more. The upper tail (P95 7,582) comes from identity errors in crowds, which is why formation agreement with v5 is 0.538 exact and 0.849 by Rand index. The analyses below are built to be robust to this error: contrasts are headlines only if their sign agrees at three pit radii, every V-layer finding has an R-layer counterpart or a sensitivity run without mostly-interpolated snapshots, and the LB-QRE fit is repeated without decisions that rest on held or interpolated positions.

3.2 Objects defined by theory

Navigation graph. We rasterise the map into 100-unit cells and call a cell walkable if a player stood on it in at least three of 1,994 archive games on the current map (2024–2026, base terrain); changing the threshold to two or five games moves median geodesic distances to the pits by 0–65 units. The walkable cells form a graph with 16-neighbourhood moves (8-neighbourhoods overestimate distances by up to 8%), and we precompute all pairwise geodesic distances. Every distance in this paper is geodesic.

Formations are set partitions. Two alive teammates are *together* if their geodesic distance is at most $r = 1{,}900$ units, five seconds of movement at the median speed. The connected components of this proximity graph partition the alive roles; with all five alive the formation is an element of the partition lattice Π_5 ordered by refinement. Π_5 has 52 elements in 7 shapes, its Hasse diagram has 160 covering edges and diameter 4 (Figure 3); all 52 partitions occur in our data. Increasing r only merges blocks, so the radius sensitivity is a chain in the lattice. Dead players are removed by restricting the partition to the alive set, never imputed.

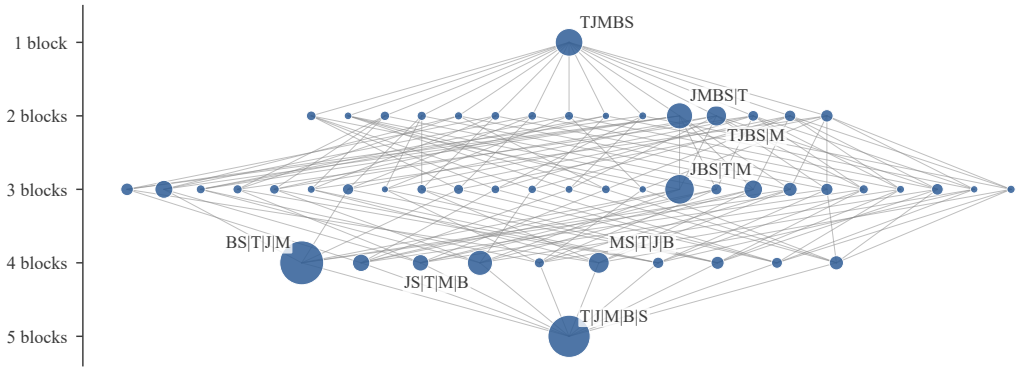


Figure 3: **The partition lattice Π_5 of the five roles (T top, J jungle, M mid, B bot, S support).** Rows are numbers of blocks; edges are covering relations (one merge). Node area is the share of V-layer team-snapshots at the spawn with all five alive.

Lemma 3.1 (Lattice metric)**.** For $\sigma, \tau \in \Pi_5$ let $d(\sigma, \tau) = |\sigma| + |\tau| - 2|\sigma \vee \tau|$, where $|\cdot|$ counts blocks and $\vee$ is the join (finest common coarsening). Then d is a metric on Π_5, equals the shortest-path length in the Hasse diagram, and every shortest path from σ to τ consists of exactly $|\sigma| - |\sigma \vee \tau|$ merges and $|\tau| - |\sigma \vee \tau|$ splits.

The metric is classical for upper semimodular lattices (Monjardet, 1981); we also verify all three statements exhaustively on Π_5 (Appendix A). It lets HASSE count merges and splits in Hasse steps, a unit that a reader can interpret.

Control and readiness. For a snapshot, each walkable cell belongs to the team whose nearest alive player reaches it first (a geodesic Voronoi split, left panel of Figure 1); *pit control* is the team's share of cells within the pit radius. The *k-arrival time* $\tau^{(k)}$ is the k-th order statistic of the team's geodesic travel times to the pit (right panel of Figure 1): the earliest moment at which the team can have k players there. k is the number of alive players within $R = 1{,}500$ units of the pit centre (sensitivity $R \in \{1{,}000, 2{,}000\}$). Each player's *zone* is one of pit, own pit side, opposite pit side, pit lane, mid, far side and base, defined relative to the pit and the team.

4 HASSE-QRE

HASSE-QRE has two layers with one rule between them. The discovery layer HASSE describes *what* setups look like and how they unfold; the explanation layer LB-QRE describes *why* teams commit the players they do. Context variables (gold, side, soul point, team strength) never enter HASSE: they are related to its output only afterwards, a modular 'cut' (Plummer, 2015). The association between patterns and context therefore cannot be built into the patterns.

4.1 HASSE: walks on the partition lattice

A team-setup e is a sequence of observations $y_{et} = (A_t, \pi_t, \{g_B\}_{B \in \pi_t})$ over $T = 13$ snapshots ($s-120$ to $s+60$ s): the alive set A_t, the formation $\pi_t \in \Pi(A_t)$, and the zone g_B of each block. A pattern c has a *centre path* $\mu_{c,1:T}$ in Π_5, a rigidity θ_c, an inertia ι_c and zone preferences ψ_c. Its formation emission is

$$P_c(\pi_t = \sigma \mid \pi_{t-1}) \propto \exp\{ -\theta_c\, d(\sigma, \mu_{c,t}|_{A_t}) - \iota_c\, d(\sigma|_C, \pi_{t-1}|_C)\}, \tag{1}$$

with d the lattice metric of Lemma 3.1, $\cdot|_A$ restriction to the alive set and $C = A_t \cap A_{t-1}$. The normaliser is an exact sum over at most 52 partitions. θ_c says how tightly the pattern holds its centre

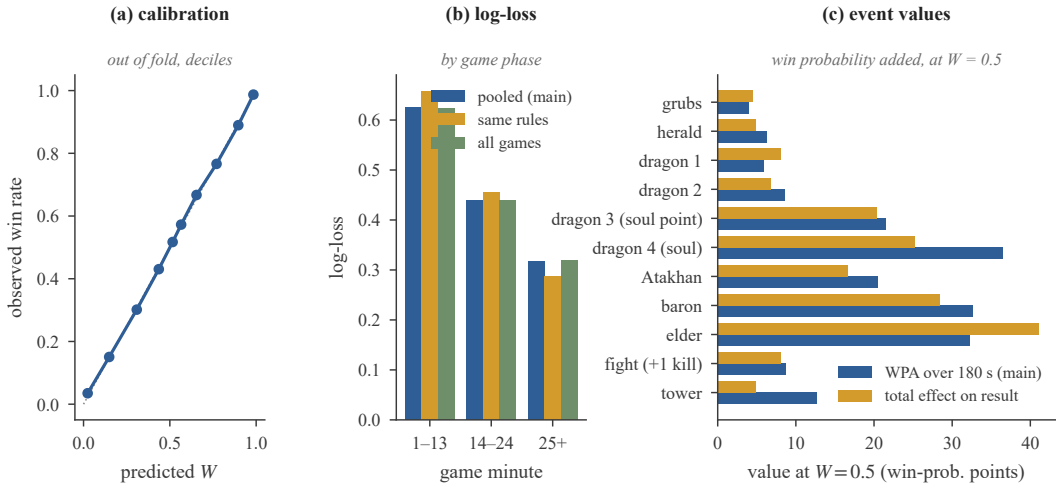


Figure 4: **Win probability prices every outcome.** (a) Calibration of the outcome-only win surface W (out of fold, deciles). (b) Log-loss by game phase for three specifications. (c) Event values at $W = 0.5$: win probability added over 180 s (main) and the total effect on the result.

and ι_c how reluctant it is to merge or split between snapshots; both are measured per Hasse step. Each block chooses a zone by a product of role experts, $P(g_B = h) \propto \exp \sum_{r \in B} \psi_c[r, h]$, with a linear time trend. Patterns follow a sticky Markov chain (Fox et al., 2011): $z_1 \sim \mathrm{Cat}(w_{\mathrm{class}(e)})$ and $P(z_t = c' \mid z_{t-1} = c) = (1 - \xi)\mathbb{1}[c' = c] + \xi q_{cc'}$; $\xi = 0$ recovers TAGNet's one-label-per-setup clustering. Snapshots after the kill are missing, not zero. We fit by EM with exact forward–backward, an exact Viterbi over the 52 centres for each μ_c, and a concave two-dimensional update for (θ_c, ι_c) (Appendix B). We examine $K = 1, \ldots, 8$ by held-out log-likelihood over five series-grouped folds, BIC and ICL (Biernacki et al., 2000); no finite K is selected (Section 5.4), so we report $K = 6$ and show $K = 5$ and $K = 8$. The restricted model H_{mono} (run at $K = 5$ with five restarts) allows the centre path only to merge, i.e. setups that only gather.

4.2 LB-QRE: THE ALLOCATION GAME

Win surface and values. $W_A(x)$ is the probability that team A wins from state x. We fit $\mathrm{logit}\, W_A(x) = \beta(t)^\top[\phi(x_A) - \phi(x_B)] + \beta_{\mathrm{side}}(t) s_A$ with $s_A = \pm 1$, which makes $W_B = 1 - W_A$ exact. ϕ contains gold (hinge basis, so curvature in gold is free), experience, kills, towers, inhibitors, plates, dragons, soul and soul point, grubs, heralds, Atakhans, barons, baron and elder buffs and alive players; $\beta(t)$ varies with the game minute through a B-spline basis, and objective coefficients are specific to the rule set. The fit uses 2,678 archive games, an L_2 penalty chosen by series-grouped cross-validation and out-of-fold predictions throughout. W is calibrated to within 1.3 pp in every decile (Figure 4a). Evaluated 15 s after the kill, W assigns the baron a *negative* value: W conditions on gold, towers and buffs, and right after the kill none of what the baron will produce has happened yet. We therefore value every event by the win probability it adds over the next 180 s, $\mathrm{logit}\, W_{t+180} = a\, \mathrm{logit}\, W_t + \beta(t)$ dir, fitted on the archive (Table 4). At $W = 0.5$ a first baron is worth 32.6 win-probability points, Atakhan 20.5, a soul-point dragon 21.6 and the first dragon 6.0; a won fight is worth 8.7 points per kill. Total effects on the final result agree (Figure 4c).

Actions. At the commitment time t_c, the first at-risk snapshot in $[s, s{+}180]$ at which some team has at least three players at the pit (otherwise the last at-risk snapshot), team i with n_i alive players has chosen $a_i = (k_i, m_i, \tau_i)$: k_i players at the pit, m_i on a second front (zone 'far'), and $\tau_i = 1$ if the pit group was already in place at the spawn. With five alive there are 36 actions; players who cannot reach the pit within 60 s are infeasible for k.

Outcomes. If both teams commit, team i wins the fight with

$$p_i = \Lambda\big(\gamma_n \ln(k_i/k_j) + \gamma_g \ln(\bar{g}_i/\bar{g}_j)\big), \tag{2}$$

the logit form of a Tullock contest (Tullock, 1980) whose numbers term generalises Lanchester's laws: under the square law with per-player strength proportional to $\bar{g}^\beta$, $\gamma_n/\gamma_g = 2/\beta$; under the linear law, $1/\beta$. The fight winner takes the objective; a lone committer takes it uncontested. The second front is a Blotto-type contest (Roberson, 2006), won with probability $m_i^{\gamma_f}/(m_i^{\gamma_f} + m_j^{\gamma_f})$ and worth a tower. γ_n and γ_g are estimated from fights in the core games (Section 5.6), outside the choice model.

Utilities and equilibrium. Team i's expected payoff splits into the objective part U_i^{obj} and the rest (fight and front) U_i^{rest}, both in win-probability points. Choices follow a heterogeneous logit quantal response equilibrium (McKelvey and Palfrey, 1995):

$$\begin{aligned}\log \sigma_i(a) = \lambda_i[\nu_i U_i^{\text{obj}}(a, \sigma_j) + U_i^{\text{rest}}(a, \sigma_j)] \\ - \kappa_i \tau(a) - \psi_i T_k(a) - \omega_{k,i} k - \omega_{m,i} m + \text{const},\end{aligned} \tag{3}$$

where λ_i is decision precision ($\lambda \to \infty$ nests Nash), ν_i the weight on the objective's own value ($\nu = 1$: the team maximises win probability), κ_i the cost of early positioning, T_k the travel minutes of the k nearest players, and ω the per-player costs of committing to the pit or the second front. Team parameters are shrunk to global values.

Proposition 4.1 (Contest threshold)**.** Suppose team i either contests, reaching win probability $W(\eta + G)$ with probability p and $W(\eta - L)$ otherwise, or concedes and trades, reaching $W(\eta - b + c)$. With logit choice at precision λ,

$$P(\text{contest} \mid \eta) = \Lambda\big(\lambda S(\eta)[p(\eta) - \pi^*(\eta)]\big),$$

$$\pi^*(\eta) = \frac{W(\eta - b + c) - W(\eta - L)}{W(\eta + G) - W(\eta - L)},$$

with stake $S(\eta) = W(\eta + G) - W(\eta - L)$.

The proposition separates the channels that the decomposition below measures. A trailing team has a lower fight probability p (Lanchester), but where W is convex its break-even π^* falls (curvature), and a valuable second front raises π^* (Blotto).

Proposition 4.2 (Uniqueness)**.** If $\nu_A = \nu_B = 1$ and costs depend only on a team's own action, the game is zero-sum in win probability up to own-action terms, and the logit QRE is the unique saddle point of

$$F(x, y) = x^\top U y - \frac{c_A^\top x - H(x)}{\lambda_A} + \frac{c_B^\top y - H(y)}{\lambda_B}$$

over the two simplices, with H the Shannon entropy.

Proofs are in Appendix A. Under the estimated $\nu \neq 1$ the game is no longer zero-sum; we solve each state by damped fixed-point iteration from uniform strategies.

Estimation and tests. We estimate $\theta_i = (\lambda_i, \nu_i, \kappa_i, \psi_i, \omega_{k,i}, \omega_{m,i})$ by nested pseudo-likelihood (Aguirregabiria and Mira, 2007) and compute intervals from 300 series-block bootstrap replicates (Künsch, 1989). Empirical content comes from fixing payoffs at outcome-only values (Haile et al., 2008); we compare the fit with a saturated multinomial benchmark. Three named tests follow from Proposition 4.1: H_{Simpson} (matched on $\hat{p}$ and stake, trailing teams contest no less than leading teams), H_{trade} (among non-committers, second-front use rises as $\hat{p}$ falls) and H_{winmax} ($\nu = 1$).

Four-channel decomposition. Let $G(S)$ be the equilibrium gap $P(k \geq 3 \mid \text{trailing}) - P(k \geq 3 \mid \text{leading})$ at the observed states when only the channels in $S \subseteq \{L, C, F, B\}$ are on: L (Lanchester; off: $\gamma_n = 0$), C (curvature; off: values taken at $W = 0.5$), F (second front; off: front values zero) and B

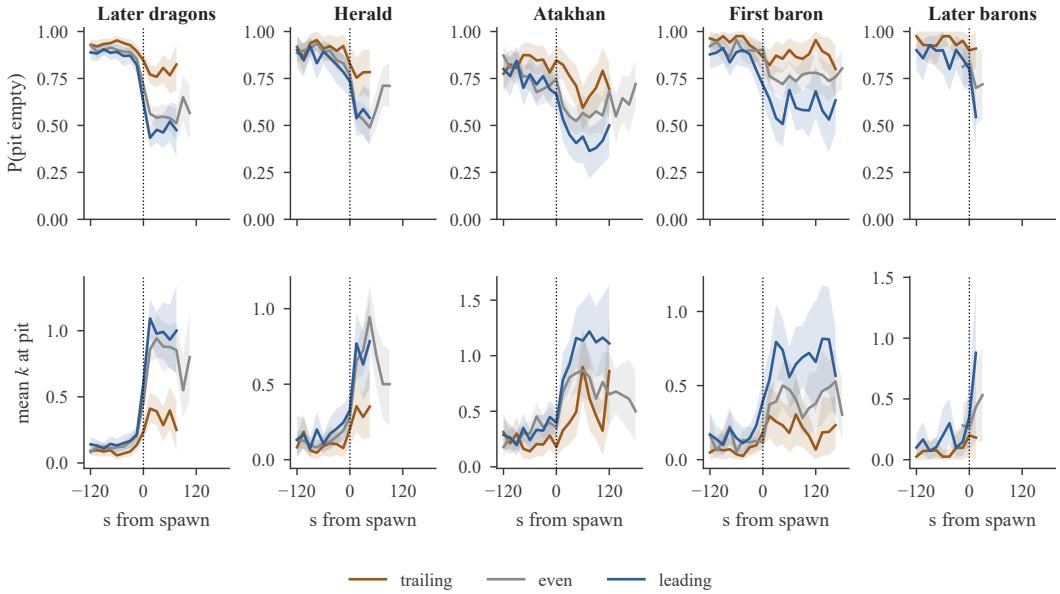


Figure 5: **Trailing teams stay away once the objective is up.** Share of team-snapshots with nobody at the pit (top) and mean number of players at the pit (bottom), by gold state, from 120 s before to 180 s after the spawn (V layer, risk set, pit radius 1,500; bands: 95% intervals; cells with fewer than 30 setups omitted).

(team behaviour; off: pooled parameters). We solve all 16 subsets and allocate $G(\{L, C, F, B\}) - G(\varnothing)$ by Shapley values (Shapley, 1953). Convexity-driven risk taking by trailing sides is known in isolation (Cabral, 2003; Skinner, 2011); the decomposition puts it in one model with the fight law and the second front and measures each in the same unit.

5 Results

Each subsection answers one question. Intervals are 95% series-block bootstrap intervals (1,000 resamples unless stated; 500 for the context model); multiple contrasts are controlled by Benjamini–Hochberg (Benjamini and Hochberg, 1995).

5.1 Trailing teams leave the pit empty

Before the spawn the three gold states look alike: at every objective most teams have nobody at the pit, and the curves overlap (Figure 5). They separate after the spawn. Leading and even teams move players in, trailing teams much less so. Figure 6 maps the trailing-minus-leading contrast over objectives and time. 53 of 300 cells at the primary radius are headlines (same sign at radii 1,000, 1,500 and 2,000; $q_{\mathrm{BH}} < 0.05$), and 53 of them point the same way: more empty pits and fewer players for trailing teams (Table 6). At the first baron the share of empty pits is +34.8 pp [18.7, 50.0] higher for trailing teams at s+45; at Atakhan +30.4 pp [15.0, 46.5], with −0.67 [−1.02, −0.32] fewer players; at respawned dragons +33.7 pp [24.9, 41.5] at s+15, where trailing teams also put three players at the pit −10.9 pp [−15.7, −5.5] less often. The only pre-spawn headline is at the herald, where trailing teams leave the pit empty +12.3 pp [3.4, 21.0] more often at s−75.

The contrast does not come from tracking error. Dropping every snapshot in which more than half of the positions are held or interpolated keeps 30 of the 53 headline cells as headlines and the sign of 42 of the cells that remain estimable; the two sets of contrasts correlate at 0.82 (Appendix Figure 18). On Riot's exact frames (R layer), the first baron at s+60 shows the same pattern: +19.7 pp [11.4, 29.9] more empty pits and −0.39 [−0.64, −0.14] players (Table 7, Figure 7a). Atakhan points the same

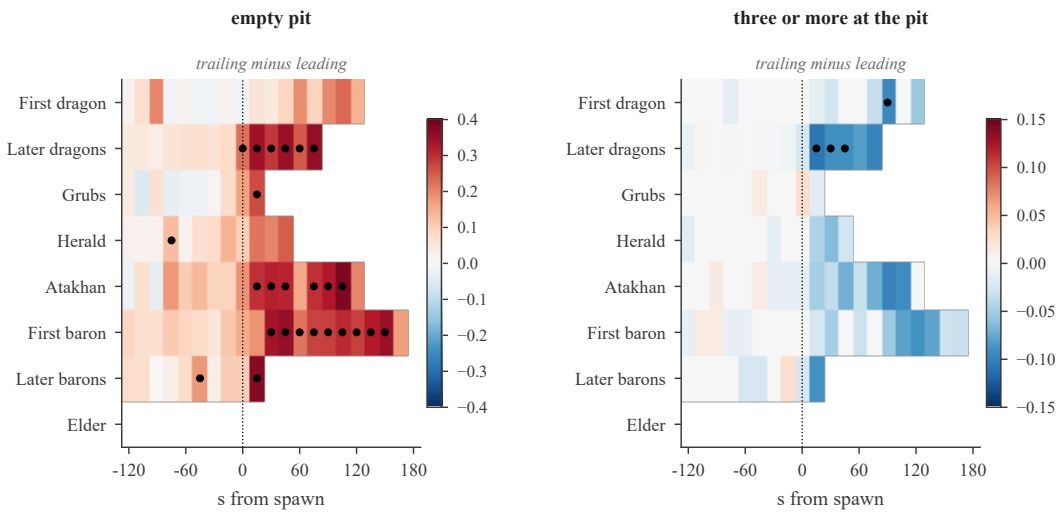


Figure 6: **The contrast is post-spawn and shared across objectives.** Trailing-minus-leading difference in the share of empty pits (left) and of three or more players at the pit (right), by objective group and time (V layer, radius 1,500). Dots mark headline cells: same sign at all three pit radii and $q_{\mathrm{BH}} < 0.05$.

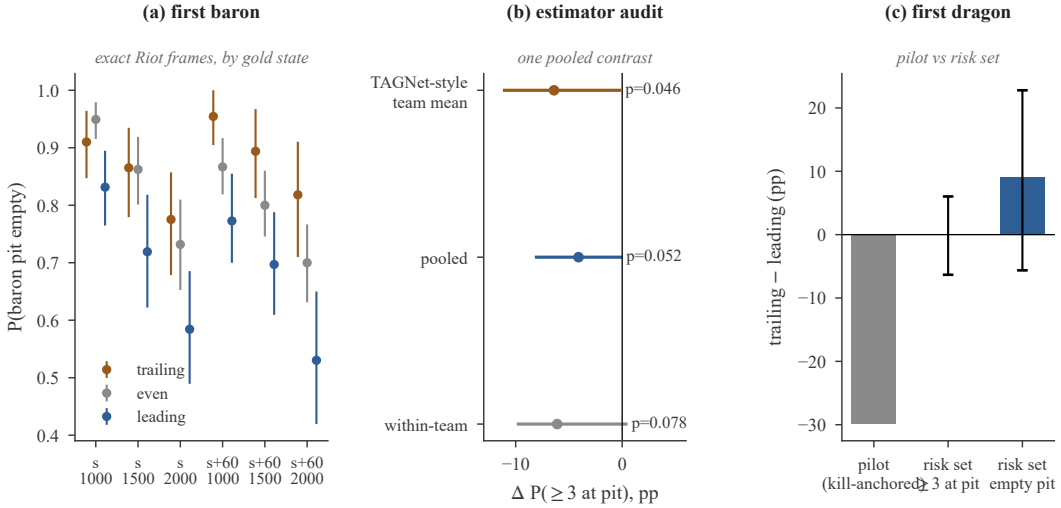


Figure 7: **Exact Riot frames confirm the baron contrast and expose two estimator artefacts.** (a) Share of empty baron pits by gold state at the spawn and 60 s later, for three pit radii. (b) One pooled contrast (share of teams with three or more at the pit at s+60, all first spawns) under three estimators; only the unweighted team-mean difference crosses $p < 0.05$ (0.046 against 0.052; both within bootstrap noise of the line). (c) The first-dragon gradient of the kill-anchored pilot against the spawn-anchored risk-set contrasts.

way on the R layer (16–22 pp across radii) with $q = 0.08$; the 60-s frames are too coarse to reach the headline bar there.

5.2 The estimator decides what is significant

The same data support different conclusions depending on the estimator (Figure 7b). Pooling the first spawns, the trailing-minus-leading difference in committing three players at s+60 is −6.4 pp [−11.2, −0.1] with $p = 0.046$ as a difference of unweighted team means (team-level averaging in the spirit of TAGNet's paired team-level test). Weighting by exposure gives −4.1 pp ($p = 0.052$), and the within-team contrast over teams observed in both states −6.1 pp ($p = 0.078$). The point estimates agree; the p-value crosses 0.05 with the estimator, although 0.046 against 0.052 lies within

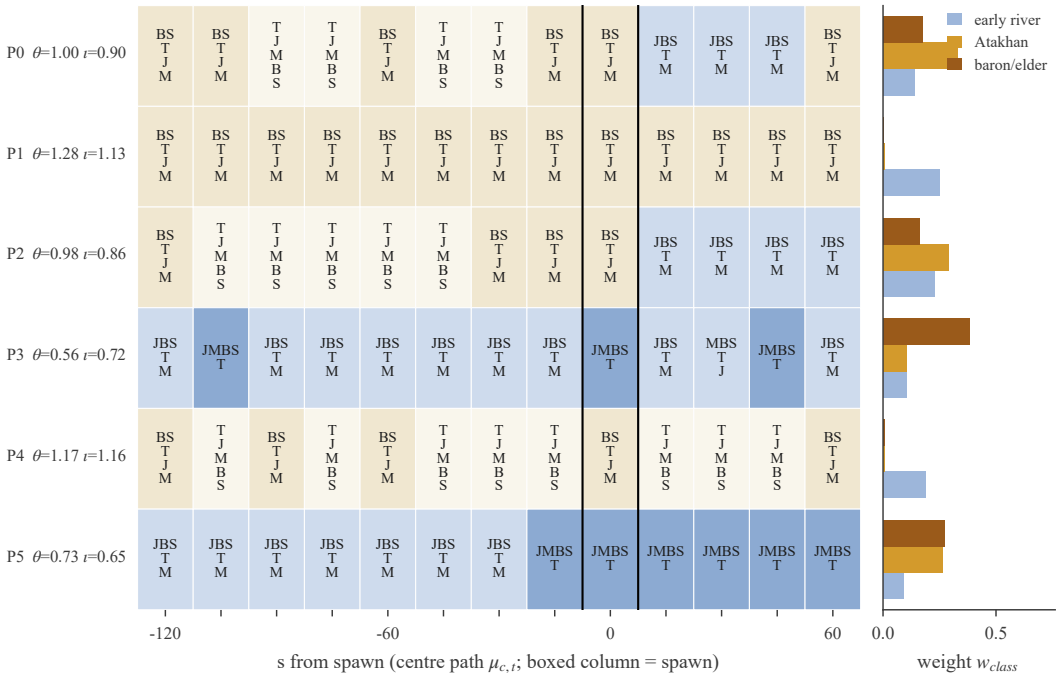


Figure 8: **The pattern dictionary** ($K = 6$)**.** Each row is a pattern's centre path $\mu_{c,t}$ on Π_5, coloured by the number of blocks (darker = more gathered), with rigidity θ and inertia ι in Hasse steps; the boxed column is the spawn. Right: mixture weight of each pattern in the three objective classes.

the bootstrap Monte-Carlo noise of about ±0.007.

Anchoring matters more. In our kill-anchored pilot on Worlds 2025, 13% of trailing and 43% of leading teams that did not take the first dragon had contested it (only 21 leading cases). Anchored at the spawn, with the risk set, the first-dragon contrast at s+60 is +9.1 pp [−5.6, 22.8] for empty pits, and committing three players is equally rare in both states (Figure 7c). At the first dragon both teams are still laning: at least 95% of pits are empty at the spawn in every gold state. The pilot gradient was produced by selecting on who took the objective.

5.3 Arriving first decides most river objectives

For each spawn we record which team first has three players within 1,500 units of the pit. That team takes the objective within 180 s in 0.80 of respawned dragons ($n = 264$), 0.77 of heralds and 0.68 of first barons, but in only 0.51 of Atakhans ($n = 106$; Figure 15). The median first arrival is 30 s after the spawn: setups are decided after the clock, not before it.

5.4 Six patterns, reproducible

No finite $K \leq 8$ is selected by any criterion on the V layer: held-out log-likelihood per setup rises from −92.85 (K=1) to −80.00 at $K = 6$ and −79.21 at $K = 8$, with steps of 0.82, 0.41 and 0.37 nats from $K = 5$ on (Figure 13, Table 8). The 7-snapshot grid and both alternative formation radii show the same rise; $K = 6$ is the largest K fitted there, and the criteria still improve at it (Appendix Table 14). We therefore report $K = 6$ as a convenient level of detail, not a selected one, and show $K = 5$ and $K = 8$ in the appendix; the rigid early-river patterns and the gathered late patterns described below appear at both.

The six patterns separate first by objective class, then by how they gather (Figure 8, Table 9). Patterns 1 and 4 carry weight 0.25 and 0.19 at early river objectives and almost none elsewhere; they keep the bot

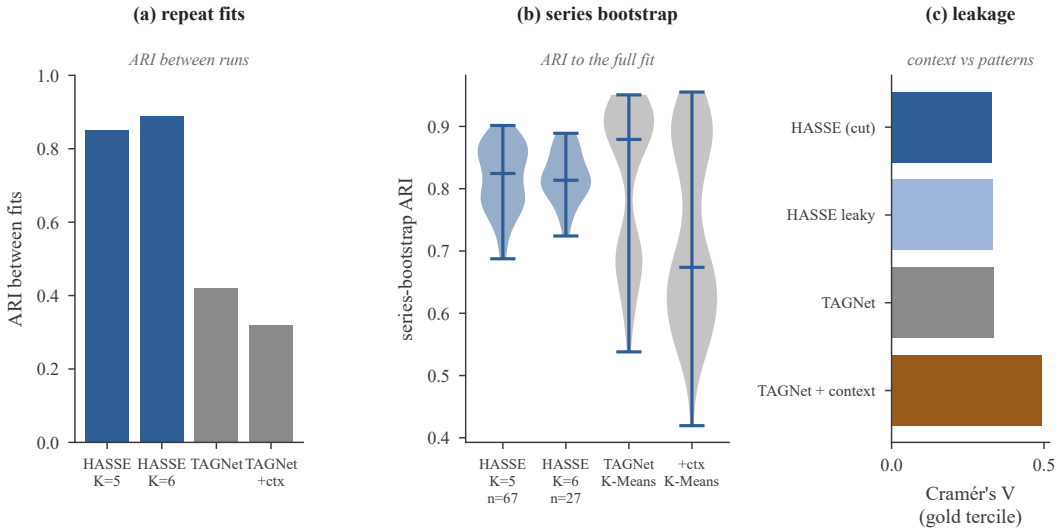


Figure 9: **Whole-model HASSE refits and TAGNet's K-Means re-runs are comparably stable, while single TAGNet runs agree less than best-of-ten HASSE fits.** (a) Agreement between repeated full-data fits: five HASSE fits, each the best of ten restarts, and ten single TAGNet runs (with and without its context vector). (b) Series-bootstrap ARI: HASSE refits the whole model on each resample, TAGNet re-runs only its K-Means step on fixed embeddings. (c) Cramér's V between the assignment and the gold tercile when context is kept out of the pattern layer and when it is fed in.

lane together and everyone else apart (`BS|T|J|M` at the spawn) and are the most rigid ($\theta = 1.28, 1.17$). Patterns 3 and 5 gather four players with the top laner apart (`JMBS|T`) at the spawn and dominate late objectives (weights 0.38 and 0.27); they are the least rigid ($\theta = 0.56, 0.73$) and the least inert ($\iota = 0.72, 0.65$): late setups change shape quickly. Patterns 0 and 2 alternate between the spread and the bot-plus-support formations before the spawn and serve Atakhan most. The model switches patterns with probability $\xi = 0.137$ per snapshot, so a setup typically keeps one pattern but can change it.

The 'only gather' model H_{mono} (at $K = 5$) fits 197 nats worse on the full data with the same number of parameters, but its held-out deficit is only 0.027 nats per setup (fold SD 0.050; 4 of 5 folds favour the unrestricted model). Splits, players peeling off to ward or flank, exist but are a small part of the structure.

Against TAGNet. We ported TAGNet to the same V-layer tracks (it uses the 2,510 setups with at least 6 of 7 pre-spawn snapshots; three-layer distance-biased graph attention, fixed GRU, K-Means at $K = 6$; ten seeds with and without its context vector). Two TAGNet runs that differ only in the seed agree at ARI (Hubert and Arabie, 1985) 0.421 (0.319 with context); five independent HASSE fits, each the best of ten restarts, agree at 0.890, so this contrast compares best-of-ten fits with single runs (Figure 9a, Table 10). Refitting the whole HASSE model on series-bootstrap resamples gives ARI 0.819 (5% quantile 0.756, 27 resamples), comparable to re-running only TAGNet's final K-Means step on fixed embeddings (0.813). TAGNet's clusters agree with its own counting baseline at ARI 0.086, and its silhouette at $K = 6$ is 0.121. Feeding the context vector in raises the association between TAGNet clusters and the gold tercile from Cramér's $V = 0.343$ to 0.425 (×1.24); the same leak into HASSE's mixture weights moves it from 0.330 to 0.331, because the weights fix only the first snapshot and the lattice emissions carry the rest.

5.5 What context does to patterns

With the patterns fixed, a soft-label multinomial logit relates occupancy to context (Figure 14). Time and objective class carry most of the variation. A one-SD gold lead raises the gathered late pattern 5 by +0.024 [0.018, 0.033] and lowers patterns 0 (−0.012) and 4 (−0.015); holding the dragon soul

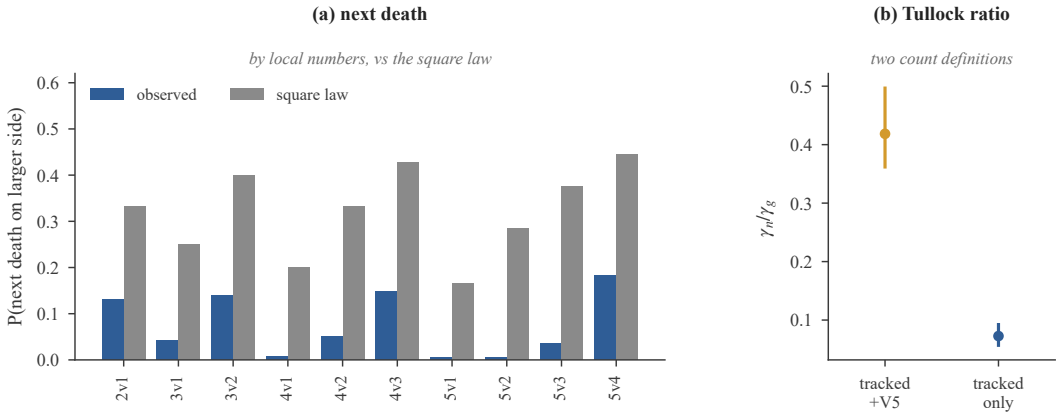


Figure 10: **The larger side rarely loses the next player.** (a) Probability that the next death falls on the larger side, by local numbers within 2,000 units, against the stochastic square law. (b) The Tullock ratio γ_n/γ_g under two count definitions.

Table 3: **Choices respond to fights, travel and player costs, and little to the objective's own value.** LB-QRE global parameters (1,454 contests, 19 teams with own parameters; 300 series-block bootstrap replicates).

	Meaning	Est.	95% CI
λ	decision precision (1/win-prob.)	4.297	[3.005, 6.457]
ν	weight on the objective's own value	0.018	[0.013, 0.026]
κ	cost of early positioning	1.794	[1.674, 1.927]
ψ	cost per travel minute	1.813	[1.476, 2.170]
ω_k	cost per player at the pit	0.185	[0.126, 0.239]
ω_m	cost per player on 2nd front	0.688	[0.639, 0.772]
ℓ	log-lik. LB-QRE / saturated	-8,343	-6,826
H_{winmax}	log-lik. loss of $\nu = 1$	62.9	[40.1, 92.8]

point raises pattern 4 by +0.030 [0.013, 0.045]. 24 of 48 effects have intervals that exclude zero.

5.6 Fight laws

The larger side almost never loses the next player (Figure 10a): 0.037 at 5v3 against 0.375 under the square law, 0.132 at 2v1 against 0.333. The slope of the next-death logit on $\ln(n_{\text{large}}/n_{\text{small}})$ is −0.57 [−0.84, −0.29] with tracked counts, between the linear law (0) and the square law (−1), and −2.33 [−2.70, −1.96] once killers and assisters from v5 are added; the levels sit far below the square law at every ratio. In the Tullock fight law (2), fitted on 1,481 fights with tracked counts, $\gamma_n = 0.79$ [0.63, 0.95] and $\gamma_g = 10.9$ [9.2, 12.8] (Table 11). With $\gamma_n/\gamma_g = 0.073$, going from 4 to 5 players is worth as much as a 1.6% per-player gold edge; side does not matter (γ_{side} =-0.029 [−0.154, 0.093]). Adding v5 participants, who are counted because they took part, raises γ_n to 4.72; we use the tracked counts and treat this as a sensitivity.

5.7 What the equilibrium reproduces

LB-QRE is fitted on 1,454 contests (Table 3). Decision precision is $\lambda = 4.30$ [3.01, 6.46] per win-probability unit, so a one-point difference in expected win probability changes choice odds by 4.4%. The weight on the objective's own value is $\nu = 0.018$ [0.013, 0.026]: choices respond to the value of fights and fronts and to their costs, early positioning ($\kappa = 1.79$), travel ($\psi = 1.81$ per minute) and committed players ($\omega_k = 0.185$, $\omega_m = 0.69$), and barely to the objective itself. Imposing $\nu = 1$ (H_{winmax}) costs 59.9 log-likelihood points on the data (likelihood ratio about 120); across bootstrap replicates the loss has mean 62.9 [40.1, 92.8] and is at least 32.8 in all 300. Without action costs the fit

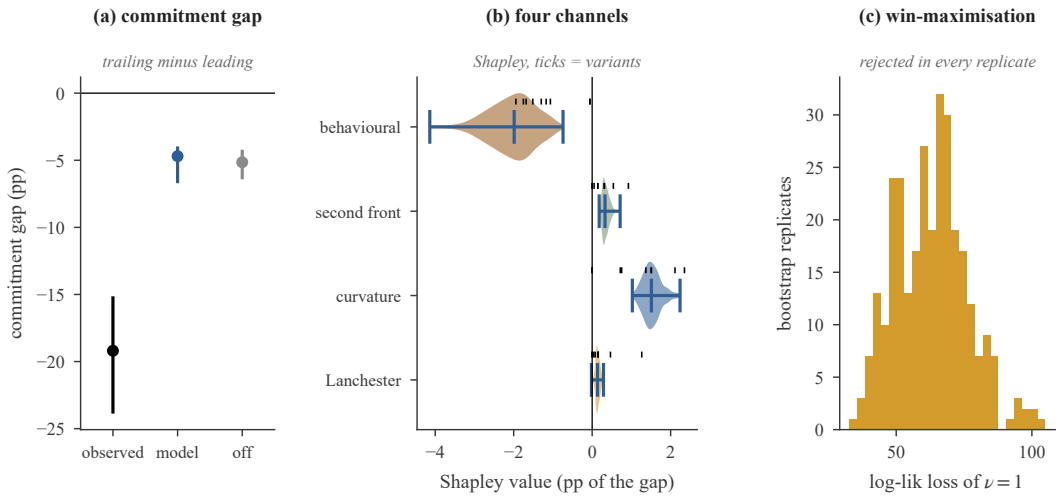


Figure 11: **The allocation game reproduces a quarter of the commitment gap, almost all of it from state variables rather than the four channels.** (a) Trailing-minus-leading gap in the share of teams with three or more players at the pit at the commitment time: observed, LB-QRE with all channels, and with all channels off. (b) Shapley values of the four channels over 300 bootstrap replicates (ticks: specification variants). (c) Log-likelihood loss from imposing $\nu = 1$ in each replicate (a nested model, so the loss is non-negative by construction; its size is the result).

collapses to $\lambda = 0$ (log-likelihood −9,254 against −8,343). The saturated benchmark reaches −6,826: the equilibrium restrictions cost likelihood, as they should if they have content. Only 57% of the 300 bootstrap replicates (170) meet the pseudo-likelihood convergence tolerance; mean ν is 0.018 in the converged and 0.019 in the other replicates, and the mean Shapley values differ by at most 0.2 pp.

The observed commitment gap is −19.2 pp [−23.9, −15.1]; the model produces −4.7 pp [−6.7, −4.0], about 0.24 of it (Figure 11a). Essentially all of the model's gap is there with every channel off (−5.1 pp), so the four channels together add only +0.5 pp net: the gold term of the fight law, the alive counts, and the travel and feasibility constraints still differ between gold states. The channels then pull in opposite directions (Table 12): curvature of W moves trailing teams towards contesting (+1.50 pp [1.14, 2.10]), team behaviour moves them away (−1.51 pp [−3.42, −1.04]), and the Lanchester (+0.15) and second-front (+0.30) channels are small. Curvature exceeds Lanchester by +1.39 pp [1.02, 1.91]. Every sign holds across the six non-degenerate specification variants (pit radius 1,000 and 2,000, result values, dropping held or interpolated decisions, the V5 fight law, $\nu = 1$), with Shapley magnitudes up to 2.4 pp; the two no-cost variants collapse to $\lambda = 0$, so their agreement is vacuous (Appendix Table 17).

The named tests sharpen the reading. Unconditionally trailing teams contest less (H_{Simpson} raw gap −0.192 [−0.237, −0.147]); matched on fight probability and stake the coefficient is +0.240 [−0.354, 0.778], in the predicted direction and not distinguishable from zero. Among teams that do not commit, second-front use does not rise as fight probability falls (H_{trade} slope +0.073 [−0.026, 0.175]). In the model, trailing teams stay away from objectives because their fights are worse and their players are farther away, not because they prefer to trade.

5.8 Team fingerprints are mostly noise

TAGNet reports team fingerprints without intervals. With intervals the picture is sober (Figure 12). Split-half correlations of team usage range from −0.04 to 0.35, and the interval includes zero for 5 of 6 patterns. Team precision parameters overlap: T1's $\lambda = 3.3$ [1.8, 6.3] and Gen.G's 3.7 [2.2, 7.3] are not distinguishable. At 31 to 394 setups per team, patterns describe the league; they do not yet separate its teams.

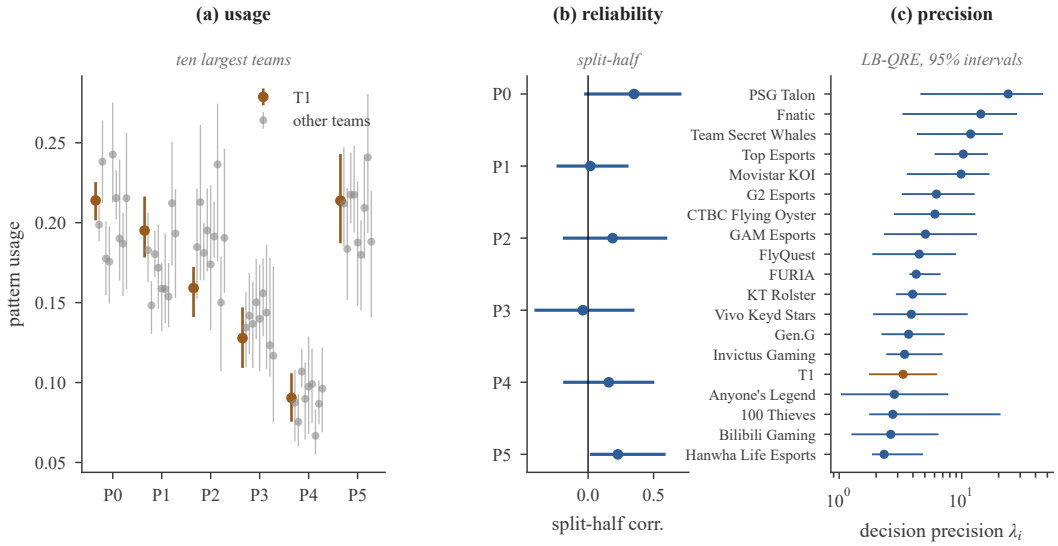


Figure 12: **Team fingerprints come with wide intervals.** (a) Pattern usage of the ten teams with most setups (T1 in red) with series-bootstrap intervals. (b) Split-half reliability of team usage per pattern. (c) Team decision precision λ_i with 95% intervals.

6 Related Work

Tactics in esports. TAGNet (Zhang et al., 2026) is the closest study and the template we mirror section by section; we replace its learned embedding by objects with units, its fixed K by held-out evidence, and its context-in-the-representation design by a cut. Spatio-temporal team behaviour in MOBAs was first studied in *Dota 2*, where skill levels differ in how teams spread across zones (Drachen et al., 2014). In *Counter-Strike*, actions have been valued by changes in win probability (Xenopoulos et al., 2020) and trajectory data released at scale (Xenopoulos and Silva, 2022). LoL research has focused on outcome prediction (Hodge et al., 2021; Yang et al., 2022), player rating (Maymin, 2021; De Bois et al., 2025), team networks (Mora-Cantallops and Sicilia, 2019) and match-level performance indicators (Novak et al., 2020). We use win probability the way action-valuation work does (Decroos et al., 2019; Xenopoulos et al., 2020), but to price the outcomes of a game between teams rather than to score individual actions.

Space and formations in team sports. Dominant regions (Taki and Hasegawa, 2000) are Voronoi cells of player positions; we compute them on a navigation graph so that walls count. Role-based alignment of formations (Lucey et al., 2013; Bialkowski et al., 2014) and the broader spatio-temporal literature (Gudmundsson and Horton, 2017) treat formations as geometric templates. We treat a formation as a set partition and use the lattice metric (Monjardet, 1981) to count merges and splits. Centred partition processes (Paganin et al., 2021) place priors around a centre partition; HASSE lets the centre move along a path over time and adds rigidity and inertia, inside a sticky HMM (Fox et al., 2011). Sparse finite mixtures (Frühwirth-Schnatter and Malsiner-Walli, 2019) motivate letting the evidence, not a fixed K, set the number of patterns.

Game theory in sport. Penalty kicks and serves are the canonical tests of equilibrium play by professionals (Walker and Wooders, 2001; Chiappori et al., 2002; Palacios-Huerta, 2003). Risk taking by trailing contestants is documented in tournaments (Genakos and Pagliero, 2012) and derived from the convexity of win probability (Cabral, 2003; Skinner, 2011), the logic behind pulling the goalie (Beaudoin and Swartz, 2010). Our explanation layer combines a quantal response equilibrium (McKelvey and Palfrey, 1995), whose empirical content requires fixed payoffs (Haile et al., 2008), with Lanchester–Tullock fights (Lanchester, 1916; Tullock, 1980) and a Blotto second front (Roberson, 2006), estimated as in empirical industrial organisation (Aguirregabiria and Mira, 2007). The Shapley decomposition (Shapley, 1953) lets the convexity argument compete with the fight law and the second

front in one unit.

7 Limitations

Positions come from the broadcast minimap; with four or more other champions nearby only 0.493 of positions are within 600 units, and formations agree exactly with Riot's in 0.538 of snapshots, which is why every headline rests on sign agreement across radii, an R-layer check or a sensitivity run without interpolated positions. Vision, cooldowns, wave states and voice communication are unobserved, so every behavioural statement is conditional on the observed state. The data cover one rule set and two international events; teams with few games carry wide intervals. LB-QRE reproduces about a quarter of the observed commitment gap, almost all of it through state variables rather than its strategic channels; the rest is behaviour the allocation game in its present form does not capture. Win probability and event values are outcome-only associations from the archive, not causal effects.

8 Conclusion

A setup before an objective is an allocation of five players on a walled map against a public clock. Measured that way, professional teams behave consistently: trailing teams leave the pit empty after the spawn, setups follow a small number of reproducible walks on the partition lattice, and commitment responds to fight odds, travel and the cost of each player far more than to the prize itself. None of this requires a neural embedding, and much of what an embedding pipeline would report changes with the seed or the estimator.

The same objects apply wherever teams split players between a contested point and the rest of the map: other MOBAs, tactical shooters and team sports with set pieces. The tracks, setups and code are released with the paper.

References

Victor Aguirregabiria and Pedro Mira. Sequential estimation of dynamic discrete games. *Econometrica*, 75(1): 1–53, 2007. doi: 10.1111/j.1468-0262.2007.00731.x.

David Beaudoin and Tim B. Swartz. Strategies for pulling the goalie in hockey. *The American Statistician*, 64(3): 197–204, 2010. doi: 10.1198/tast.2010.09147.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

Alina Bialkowski, Patrick Lucey, Peter Carr, Yisong Yue, Sridha Sridharan, and Iain Matthews. Large-scale analysis of soccer matches using spatiotemporal tracking data. In *IEEE International Conference on Data Mining*, pages 725–730, 2014. doi: 10.1109/ICDM.2014.133.

Christophe Biernacki, Gilles Celeux, and Gérard Govaert. Assessing a mixture model for clustering with the integrated completed likelihood. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(7): 719–725, 2000. doi: 10.1109/34.865189.

Luís M. B. Cabral. R&D competition when firms choose variance. *Journal of Economics & Management Strategy*, 12(1):139–150, 2003. doi: 10.1162/105864003321220760.

Pierre-André Chiappori, Steven Levitt, and Timothy Groseclose. Testing mixed-strategy equilibria when players are heterogeneous: The case of penalty kicks in soccer. *American Economic Review*, 92(4):1138–1151, 2002. doi: 10.1257/00028280260344678.

Maxime De Bois, Flora Parmentier, Raphaël Puget, Matthew Tanti, and Jordan Peltier. PandaSkill—player performance and skill rating in esports: Application to League of Legends. *IEEE Transactions on Games*, 17 (4):989–1002, 2025. doi: 10.1109/TG.2025.3581070.

Tom Decroos, Lotte Bransen, Jan Van Haaren, and Jesse Davis. Actions speak louder than goals: Valuing player actions in soccer. In *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery & Data Mining*, pages 1851–1861, 2019. doi: 10.1145/3292500.3330758.

Anders Drachen, Matthew Yancey, John Maguire, Derrek Chu, Iris Yuhui Wang, Tobias Mahlmann, Matthias Schubert, and Diego Klabajan. Skill-based differences in spatio-temporal team behavior in Defence of The Ancients 2. In *IEEE Games, Entertainment, Media Conference (GEM)*, 2014.

Emily B. Fox, Erik B. Sudderth, Michael I. Jordan, and Alan S. Willsky. A sticky HDP-HMM with application to speaker diarization. *The Annals of Applied Statistics*, 5(2A):1020–1056, 2011. doi: 10.1214/10-AOAS395.

Sylvia Frühwirth-Schnatter and Gertraud Malsiner-Walli. From here to infinity: sparse finite versus Dirichlet process mixtures in model-based clustering. *Advances in Data Analysis and Classification*, 13(1):33–64, 2019. doi: 10.1007/s11634-018-0329-y.

Christos Genakos and Mario Pagliero. Interim rank, risk taking, and performance in dynamic tournaments. *Journal of Political Economy*, 120(4):782–813, 2012. doi: 10.1086/668502.

Joachim Gudmundsson and Michael Horton. Spatio-temporal analysis of team sports. *ACM Computing Surveys*, 50(2):1–34, 2017. doi: 10.1145/3054132.

Philip A. Haile, Ali Hortaçsu, and Grigory Kosenok. On the empirical content of quantal response equilibrium. *American Economic Review*, 98(1):180–200, 2008. doi: 10.1257/aer.98.1.180.

Victoria J. Hodge, Sam Devlin, Nick Sephton, Florian Block, Peter I. Cowling, and Anders Drachen. Win prediction in multiplayer esports: Live professional match prediction. *IEEE Transactions on Games*, 13(4): 368–379, 2021. doi: 10.1109/TG.2019.2948469.

Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi: 10.1007/BF01908075.

Hans R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17 (3):1217–1241, 1989. doi: 10.1214/aos/1176347265.

Frederick William Lanchester. *Aircraft in Warfare: The Dawn of the Fourth Arm*. Constable, London, 1916.

Leaguepedia. Leaguepedia: League of Legends esports wiki. https://lol.fandom.com. Accessed 2026-10.

Patrick Lucey, Alina Bialkowski, Peter Carr, Stuart Morgan, Iain Matthews, and Yaser Sheikh. Representing and discovering adversarial team behaviors using player roles. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 2706–2713, 2013. doi: 10.1109/CVPR.2013.349.

Philip Z. Maymin. Smart kills and worthless deaths: eSports analytics for League of Legends. *Journal of Quantitative Analysis in Sports*, 17(1):11–27, 2021. doi: 10.1515/jqas-2019-0096.

Richard D. McKelvey and Thomas R. Palfrey. Quantal response equilibria for normal form games. *Games and Economic Behavior*, 10(1):6–38, 1995. doi: 10.1006/game.1995.1023.

Bernard Monjardet. Metrics on partially ordered sets—a survey. *Discrete Mathematics*, 35(1-3):173–184, 1981. doi: 10.1016/0012-365X(81)90206-5.

Marçal Mora-Cantallops and Miguel-Ángel Sicilia. Team efficiency and network structure: The case of professional League of Legends. *Social Networks*, 58:105–115, 2019. doi: 10.1016/j.socnet.2019.03.004.

Andrew R. Novak, Kyle J. M. Bennett, Matthew A. Pluss, and Job Fransen. Performance analysis in esports: modelling performance at the 2018 League of Legends world championship. *International Journal of Sports Science & Coaching*, 15(5-6):809–817, 2020. doi: 10.1177/1747954120932853.

Sally Paganin, Amy H. Herring, Andrew F. Olshan, and David B. Dunson. Centered partition processes: Informative priors for clustering (with discussion). *Bayesian Analysis*, 16(1):301–370, 2021. doi: 10.1214/20-BA1197.

Ignacio Palacios-Huerta. Professionals play minimax. *Review of Economic Studies*, 70(2):395–415, 2003. doi: 10.1111/1467-937X.00249.

Martyn Plummer. Cuts in Bayesian graphical models. *Statistics and Computing*, 25(1):37–43, 2015. doi: 10.1007/s11222-014-9503-z.

Riot Games. Riot developer portal: MATCH-V5 API. https://developer.riotgames.com/apis. Accessed 2026-10.

Brian Roberson. The Colonel Blotto game. *Economic Theory*, 29(1):1–24, 2006. doi: 10.1007/s00199-005-0071-5.

Lloyd S. Shapley. A value for n-person games. In *Contributions to the Theory of Games, Volume II*, Annals of Mathematics Studies 28, pages 307–317. Princeton University Press, 1953. doi: 10.1515/9781400881970-018.

Brian Skinner. Scoring strategies for the underdog: A general, quantitative method for determining optimal sports strategies. *Journal of Quantitative Analysis in Sports*, 7(4), 2011. doi: 10.2202/1559-0410.1364.

Tsuyoshi Taki and Jun-ichi Hasegawa. Visualization of dominant region in team games and its application to teamwork analysis. In *Proceedings Computer Graphics International*, pages 227–235, 2000. doi: 10.1109/CGI.2000.852338.

Gordon Tullock. Efficient rent seeking. In James M. Buchanan, Robert D. Tollison, and Gordon Tullock, editors, *Toward a Theory of the Rent-Seeking Society*, pages 97–112. Texas A&M University Press, College Station, TX, 1980.

Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations (ICLR)*, 2018.

Mark Walker and John Wooders. Minimax play at Wimbledon. *American Economic Review*, 91(5):1521–1538, 2001. doi: 10.1257/aer.91.5.1521.

Peter Xenopoulos and Claudio Silva. ESTA: An esports trajectory and action dataset, 2022.

Peter Xenopoulos, Harish Doraiswamy, and Claudio Silva. Valuing player actions in Counter-Strike: Global Offensive. In *IEEE International Conference on Big Data*, 2020.

Zelong Yang, Yan Wang, Piji Li, Shaobin Lin, Shuming Shi, Shao-Lun Huang, and Wei Bi. Predicting events in MOBA games: Prediction, attribution, and evaluation. *IEEE Transactions on Games*, 2022.

Xiang Zhang, RongQi Zhang, Guancheng Chen, Xinlei Cao, ChaoFan Yue, and Chunhong Yuan. TAGNet: Temporal attention graph network for defensive tactical pattern recognition in professional Counter-Strike 2. *IEEE Transactions on Games*, pages 1–13, 2026. doi: 10.1109/TG.2026.3745970. Early Access.

A Proofs

Proof of Lemma 3.1. Π_5 ordered by refinement is a geometric, hence upper semimodular, lattice with rank $r(\pi) = 5 - |\pi|$. For upper semimodular lattices $d(\sigma, \tau) = 2r(\sigma \vee \tau) - r(\sigma) - r(\tau)$ is a metric equal to the shortest-path distance in the covering graph (Monjardet, 1981); substituting the rank gives the stated form. Each covering step is one merge (rank +1) or one split (rank −1). On a path from σ to τ with M merges and S splits, $M - S = r(\tau) - r(\sigma)$; on a shortest path also $M + S = d(\sigma, \tau) = 2r(\sigma \vee \tau) - r(\sigma) - r(\tau)$. Hence $M = r(\sigma \vee \tau) - r(\sigma) = |\sigma| - |\sigma \vee \tau|$ and $S = |\tau| - |\sigma \vee \tau|$. We verified all three statements exhaustively: breadth-first search on the Hasse diagram of the 52 partitions (160 covering edges) reproduces d for all pairs, and the triangle inequality holds for all triples. ■

Proof of Proposition 4.1. Contesting yields expected win probability $pW(\eta + G) + (1 - p)W(\eta - L)$, conceding yields $W(\eta - b + c)$. Their difference is $W(\eta - L) + pS - W(\eta - b + c) = S(p - \pi^*)$ with S and π^* as stated. Under logit choice with precision λ the probability of the first option is $\Lambda(\lambda \cdot \text{difference})$. ■

Proof of Proposition 4.2. With $\nu = 1$, $W_B = 1 - W_A$ implies that every outcome changes B's win probability by minus A's change, so $U_B = -U_A^\top$ in win-probability units, and the remaining terms ($\kappa\tau$, ψT_k, ω) depend only on a team's own action; write them $c_A^\top x$ and $c_B^\top y$ for mixed strategies x, y. Team A's logit best response to y maximises $\lambda_A x^\top U y - c_A^\top x + H(x)$, team B's maximises $-\lambda_B x^\top U y - c_B^\top y + H(y)$. Dividing by λ_A and λ_B, both are the first-order conditions of the saddle problem $\max_x \min_y F(x, y)$ with F as stated. F is continuous, strictly concave in x and strictly convex in y on compact convex sets, so a saddle point exists (minimax theorem) and is unique by strictness; every logit QRE is such a saddle point and conversely. ■

B HASSE: Estimation

Likelihood. For setup e with pattern sequence $z_{1:T}$, the complete-data likelihood is $w_{\mathrm{cls}(e),z_1} \prod_t P(z_t \mid z_{t-1}) \prod_t P_{z_t}(\pi_t \mid \pi_{t-1}) \prod_{B \in \pi_t} P_{z_t}(g_B \mid \pi_t)$; snapshots after the kill are missing and drop out of the product.

E step. Exact forward–backward over K states.

M step. (i) For each pattern and snapshot, the centre $\mu_{c,t}$ is chosen among the 52 partitions by an exact Viterbi pass that adds a smoothness penalty of one unit per Hasse step between consecutive centres. (ii) (θ_c, ι_c) maximise a concave exponential-family likelihood whose gradient is observed minus expected distance, bounded in $[0, 20]$. (iii) Zone parameters by penalised multinomial logit (Gaussian penalty 0.1). (iv) w, ξ, q in closed form.

Initialisation and selection. Ten restarts per K, even seeds from k-medoids on the lattice path distance; held-out log-likelihood on five series-grouped folds; BIC and ICL on the full data. Recovery on synthetic data drawn from the model (three planted patterns, 600 setups, three snapshots, five restarts): all restarts reach the same log-likelihood (spread 0.0), 3 of 3 centre paths are recovered exactly, rigidity and inertia lie within 0.10 and 0.08 of the planted values, and the first-snapshot assignment has ARI 0.915 with the truth (0.903 when the last snapshot is missing in 40% of setups).

H_{mono}**.** The Viterbi step for μ is restricted to transitions $\mu_t \leq \mu_{t+1}$ in the refinement order.

C LB-QRE: Specification

Commitment time and actions. t_c is the first at-risk snapshot in $[s, s{+}180]$ with three or more players of either team within R of the pit, otherwise the last at-risk snapshot (objective taken next, or

Table 4: **Every event is priced in win-probability points.** Event values in win-probability points at $W = 0.5$, from outcome-only regressions on the archive (logit $W_{t+180} = a \operatorname{logit} W_t + \beta(t)$ dir); 'total effect' replaces W_{t+180} by the result.

Event	n	a	WPA (180 s)	Total effect
Void grubs	518	1.00	4.0	4.5
Rift Herald	489	1.00	6.2	4.9
Dragon 1	4,612	1.03	6.0	8.1
Dragon 2	3,544	1.01	8.5	6.8
Dragon 3 (soul point)	2,395	0.85	21.6	20.3
Dragon 4 (soul)	1,145	0.53	36.4	25.2
Atakhan	487	0.95	20.5	16.7
Baron	523	0.71	32.6	28.5
Elder	257	0.74	32.2	41.1
Fight, per kill (away from objectives)	14,563	0.84	8.7	8.1
Fight, per kill (near objective)	18,891	0.88	8.4	8.7
Tower	31,090	0.81	12.7	4.9

Table 5: **Median tracking error is small against the map width and grows with crowding.** Tracker error against Riot V5 positions at held-out odd minutes (23,286 champion-minutes, 164 games; map units, map width 14,820). Crowding = other champions within 2,000 units.

Subset	%	Median	P90	<300	<600	<1000
pooled	100	177	3,739	0.70	0.79	0.82
detected	78	146	589	0.83	0.90	0.91
interpolated	21	921	5,894	0.25	0.40	0.52
held	1	3,012	7,151	0.06	0.10	0.15
crowding 0	29	146	502	0.82	0.90	0.92
crowding 1	28	148	1,374	0.81	0.88	0.89
crowding 2	13	180	3,269	0.71	0.81	0.83
crowding 3	11	214	6,061	0.61	0.72	0.76
crowding 4+	19	621	7,675	0.39	0.49	0.58

window end). m counts alive players in zone 'far' at t_c; $\tau = 1$ if the team already had at least k players at the pit at the spawn. Players farther than 60 s of movement at the spawn are infeasible for the pit.

State values. For each contest, values for the objective (to either team), a fight won or lost by two kills, and one tower are read from the event regressions at the out-of-fold W of the last v5 frame at least 30 s before the spawn; dragons resolve to the gainer's next drake index.

Estimation. Nested pseudo-likelihood alternates a conditional-logit maximisation in θ (analytic gradient) with a strategy update; team deviations are shrunk towards global values with prior standard deviations 1.0 for $\log \lambda$, κ, ψ and 0.5 for $\log \nu$, ω_k, ω_m. Teams with too few contests share a pooled 'other' parameter.

D Additional Results

E T1 and Faker

T1 contributes 394 setups, the most of any team, and won Worlds 2025. Every analysis here was specified before it was run, compares T1 or Faker with the rest of the league on the same measures, and reports intervals; null results are reported as such.

Table 6: **Headline contrasts concentrate after the spawn.** V layer: headline trailing-minus-leading contrasts (sign agrees at three pit radii, $q_{BH} < 0.05$). Cells = number of headline time points; the largest is shown. Units: pp for shares, players for mean k.

Objective	Quantity	Cells	Peak at	T−L	95% CI
Herald	pit empty	1	$s - 75$	+12.3	[3.4, 21.0]
Herald	mean k	1	$s - 75$	−0.15	[−0.27, −0.05]
Grubs	pit empty	1	$s + 15$	+26.2	[6.9, 46.7]
Grubs	mean k	1	$s + 15$	−0.36	[−0.64, −0.12]
First dragon	≥3 at pit	1	$s + 90$	−9.6	[−17.0, −2.2]
First dragon	mean k	1	$s + 105$	−0.39	[−0.66, −0.11]
Later dragons	pit empty	6	$s + 75$	+35.0	[19.0, 51.5]
Later dragons	≥3 at pit	3	$s + 15$	−10.9	[−15.7, −5.5]
Later dragons	mean k	7	$s + 75$	−0.75	[−1.05, −0.42]
Atakhan	pit empty	6	$s + 105$	+37.2	[20.0, 54.0]
Atakhan	mean k	5	$s + 105$	−0.84	[−1.28, −0.43]
First baron	pit empty	9	$s + 45$	+34.8	[18.7, 50.0]
First baron	mean k	8	$s + 135$	−0.63	[−1.03, −0.22]
Later barons	pit empty	2	$s + 15$	+36.4	[15.4, 59.5]
Later barons	mean k	1	$s - 45$	−0.27	[−0.45, −0.12]

Table 7: **Exact Riot frames confirm the baron contrast and the estimator dependence.** R layer (exact Riot frames) at s+60, risk set, pit radius 1,500. Top: trailing minus leading by objective (headline = sign agrees at radii 1,000/1,500/2,000 and $q_{BH} < 0.05$). Bottom: one pooled contrast under three estimators.

Objective / estimator	Quantity	T−L (pp)	95% CI	q_{BH}	Head.
Elemental dragon	pit empty	+9.1	[−5.6, 22.8]	0.584	
	mean k	−0.13	[−0.43, 0.15]	0.769	
Atakhan	pit empty	+16.4	[1.4, 32.3]	0.213	
	mean k	−0.37	[−0.78, 0.00]	0.269	
Baron Nashor	pit empty	+19.7	[11.4, 29.9]	0.000	✓
	mean k	−0.39	[−0.64, −0.14]	0.000	✓
TAGNet-style team mean	≥3 at pit	−6.4	[−11.2, −0.1]	p=0.046	
pooled (exposure-weighted)	≥3 at pit	−4.1	[−8.2, 0.0]	p=0.052	
within-team (teams seen in both states)	≥3 at pit	−6.1	[−9.9, 0.5]	p=0.078	

A1 Pattern fingerprint. T1's usage over the six patterns is 0.21/0.20/0.16/0.13/0.09/0.21 (Table 19). Its nearest team is Gen.G (total-variation distance 0.032; nearest in 40% of resamples), then Top Esports and KT Rolster. Given the low split-half reliability of all fingerprints (Section 5.8), this places T1 inside the league's main cluster rather than apart from it.

A2 Decision parameters. T1's precision is $\lambda = 3.3$ [1.8, 6.3] and its precision on the objective $\lambda\nu = 0.060$ [0.032, 0.145]; neither separates T1 from Gen.G ($\lambda = 3.7$ [2.2, 7.3]) or from the league (Figure 26, Table 18).

A3 Faker in the setup. Against other starting mid laners ($n = 432$ vs 2,545 setups), Faker is closer to the pit before the spawn by −469 [−709, −275] units and arrives earlier in his team's order (rank −0.12 [−0.23, −0.03]); he is no more often alone, in the main block or central in the team's spanning tree (Table 20, Figure 27a). 'Faker gets to the objective first' is measurable and true; 'Faker holds the formation together' is measurable and not supported.

A5 Behind and ahead. T1's trailing-minus-leading commitment contrast is −0.194 [−0.320, −0.064], the rest of the league's −0.192; the difference is −0.002 [−0.133, 0.149] (Table 21). 'T1 does not panic when behind' describes the whole league as well as T1.

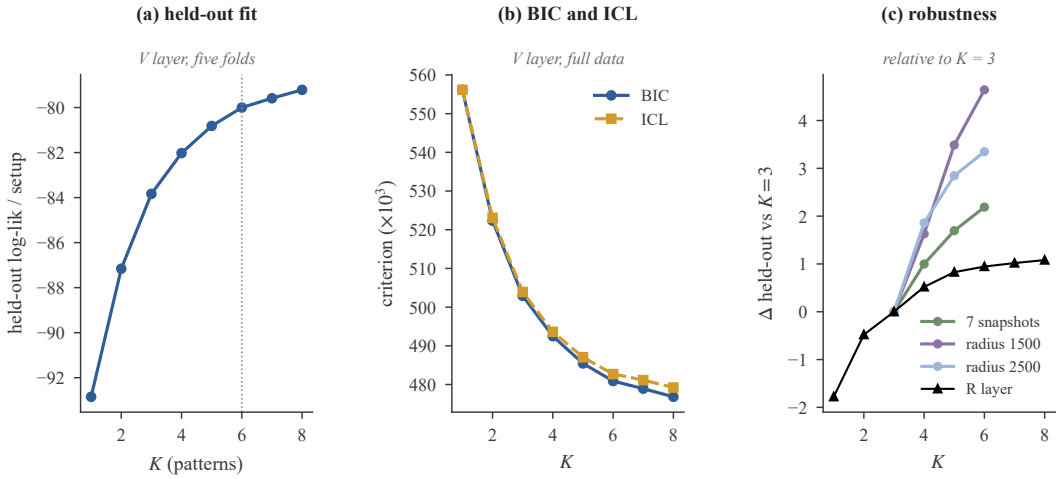


Figure 13: **The evidence favours rich structure with diminishing returns.** (a) Held-out log-likelihood per setup over five series-grouped folds against the number of patterns K (V layer). (b) BIC and ICL. (c) Held-out log-likelihood relative to $K = 3$ for the 7-snapshot grid, formation radii 1,500 and 2,500, and the R layer.

Table 8: **No finite $K \leq 8$ is selected on the V layer.** HASSE model selection. V layer (13 snapshots, 2,994 setups) and R layer (3 exact frames); held-out log-likelihood per setup over five series-grouped folds.

K	Params	Held-out/setup	BIC	ICL	R: held-out	R: BIC
1	76	-92.85	556,150	556,150	-22.48	73,517
2	154	-87.16	522,326	523,092	-21.18	69,614
3	235	-83.83	502,918	503,899	-20.70	68,225
4	318	-82.02	492,481	493,618	-20.18	66,849
5	403	-80.82	485,443	487,048	-19.87	66,306
6	490	-80.00	480,902	482,704	-19.75	66,484
7	579	-79.59	478,926	481,145	-19.68	66,689
8	670	-79.21	476,828	479,225	-19.62	66,798

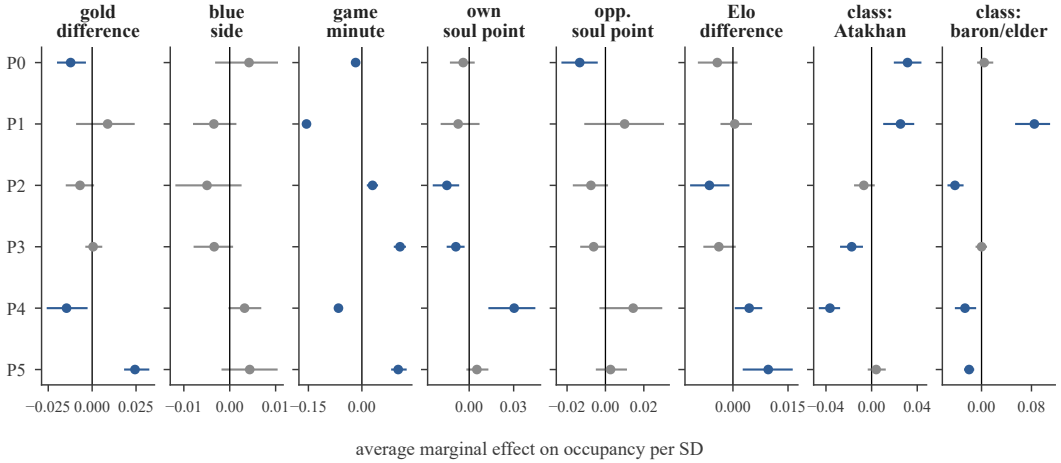


Figure 14: **Context shifts pattern use modestly once time and objective class are held fixed.** Average marginal effect on each pattern's occupancy per standard deviation of each covariate, from a soft-label multinomial logit fitted after the patterns (cut design); blue: 95% interval excludes zero.

A6 Fights. In 399 fights involving T1, the mean residual of the fight law is −0.002 [−0.045, 0.046]: at equal numbers and gold T1 wins its fights as often as the law predicts.

Table 9: **Six patterns separate by objective class and by how they gather.** The pattern dictionary ($K = 6$, V layer). Centre path $\mu_{c,t}$ at four of the 13 snapshots (T top, J jungle, M mid, B bot, S support; bars separate blocks), rigidity θ and inertia ι (per Hasse step) and the mixture weight in each objective class. Switching probability $\xi = 0.137$.

	θ	ι	w_{river}	$w_{Atak.}$	w_{late}	s−90	s−30	s	s+60
P0	1.00	0.90	0.14	0.33	0.18	T|J|M|B|S	T|J|M|B|S	BS|T|J|M	BS|T|J|M
P1	1.28	1.13	0.25	0.01	0.00	BS|T|J|M	BS|T|J|M	BS|T|J|M	BS|T|J|M
P2	0.98	0.86	0.23	0.29	0.16	T|J|M|B|S	BS|T|J|M	BS|T|J|M	JBS|T|M
P3	0.56	0.72	0.10	0.11	0.38	JBS|T|M	JBS|T|M	JMBS|T	JBS|T|M
P4	1.17	1.16	0.19	0.01	0.01	BS|T|J|M	T|J|M|B|S	BS|T|J|M	BS|T|J|M
P5	0.73	0.65	0.09	0.26	0.27	JBS|T|M	JBS|T|M	JMBS|T	JMBS|T

Table 10: **HASSE refits agree where TAGNet seeds do not.** HASSE against TAGNet ported to the same V-layer tracks (TAGNet uses the 2,510 setups with at least 6 of 7 pre-spawn snapshots). HASSE rows are five independent fits, each the best of ten restarts; TAGNet rows are ten single runs per variant, with and without its context vector.

	HASSE ($K = 6$)	TAGNet port
Setups used	2,994 (all)	2,510 (≥6 of 7 pre-spawn)
Choice of K	held-out lik., BIC, ICL	fixed $K = 6$ (K-Means)
Repeat-fit ARI	0.890 (restarts)	0.421 / 0.319 (seeds)
Series-bootstrap ARI	0.819 (q05 0.756; whole model)	0.813 (q05 0.608; K-Means only)
Agreement with counting baseline	–	ARI 0.086
Cramér's V with gold (clean)	0.330	0.337
Cramér's V with gold (context fed in)	0.331	0.494 (×1.24)
Parameters with units	θ, ι (Hasse steps), ξ, zones	64-d embedding
'only gather' test H_{mono}	Δ held-out 0.027/setup (4/5 folds)	–

A7 The Worlds 2025 final. Table 22 and Figure 28 list all 51 spawns of T1's 3–2 final against KT Rolster with both actions, the fight probability, the break-even of Proposition 4.1 and the objective's value.

A8 MSI and Worlds. Under the same rules T1 committed three or more players in 0.203 of contests at MSI and 0.254 at Worlds (Figure 27c).

A9 Earlier rule sets. Worlds 2023 and 2024 were played under different objective rules, so T1's earlier setups are outside this study.

A10 Narratives and measures. Three common narratives map onto measures here: 'T1 does not panic when behind' (A5: no different from the league), 'Faker controls the map' (A3: earlier arrival, no extra centrality) and 'T1 wins fights at equal numbers' (A6: no residual). Champion-pool depth, draft play and communication are outside what positions can measure.

F Dataset

The release contains, for the 164 core games: the clock alignment per video, minimap registration parameters, 2-fps tracks with a source label per position (detected, interpolated, held, pinned, dead), the 1,516 spawn records, the 2,994 team-setups on the 15-s grid with partitions, zones, k, $\tau^{(k)}$ and pit control and the navigation graph; the code that produces every figure and table is released with them.

Table 11: **The larger side loses fewer players than the square law predicts.** Fight laws. Next death: logit of the next death falling on the larger side on $\ln(n_{large}/n_{small})$ and gold; Tullock: fight win on $\ln(k_i/k_j)$ and $\ln(\bar{g}_i/\bar{g}_j)$. Counts within 2,000 units 1 s before a kill; 'tracked+V5' adds V5 killers and assisters (endogenous; sensitivity).

Model	Counts	Parameter	Est.	95% CI	n
Next death	tracked+V5	intercept	−0.945	[−1.162, −0.715]	3,877
Next death	tracked+V5	b_n (square −1, linear 0)	−2.327	[−2.700, −1.959]	3,877
Next death	tracked+V5	b_g (gold)	−4.652	[−5.901, −3.430]	3,877
Tullock fight	tracked+V5	γ_n (numbers)	4.719	[4.245, 5.272]	2,139
Tullock fight	tracked+V5	γ_g (gold)	11.276	[9.636, 13.097]	2,139
Tullock fight	tracked+V5	γ_{side}	−0.059	[−0.199, 0.080]	2,139
Tullock fight	tracked+V5	γ_n/γ_g	0.419	[0.359, 0.499]	2,139
Next death	tracked	intercept	−0.188	[−0.400, 0.019]	2,683
Next death	tracked	b_n (square −1, linear 0)	−0.566	[−0.839, −0.286]	2,683
Next death	tracked	b_g (gold)	−7.034	[−8.138, −6.002]	2,683
Tullock fight	tracked	γ_n (numbers)	0.795	[0.630, 0.952]	1,481
Tullock fight	tracked	γ_g (gold)	10.936	[9.205, 12.830]	1,481
Tullock fight	tracked	γ_{side}	−0.029	[−0.154, 0.093]	1,481
Tullock fight	tracked	γ_n/γ_g	0.073	[0.054, 0.095]	1,481

Table 12: **Curvature pulls trailing teams towards contesting; team behaviour pulls them away.** Decomposition of the trailing-minus-leading commitment gap (pp) and the named tests (theory direction in brackets).

Quantity	Est.	95% CI	$P(>0)$
observed gap	−19.19	[−23.88, −15.14]	0.00
model gap (all channels)	−4.70	[−6.71, −3.97]	0.00
model gap (channels off)	−5.15	[−6.42, −4.22]	0.00
Shapley: Lanchester	+0.15	[0.05, 0.25]	1.00
Shapley: curvature of W	+1.50	[1.14, 2.10]	1.00
Shapley: second front	+0.30	[0.21, 0.58]	1.00
Shapley: behavioural	−1.51	[−3.42, −1.04]	0.00
$H_{Simpson}$ raw gap [< 0 (trailing contest less)]	−0.192	[−0.237, −0.147]	p=0.000
$H_{Simpson}$ matched (logit) [≥ 0]	+0.240	[−0.354, 0.778]	p=0.464
H_{trade} slope [< 0]	+0.073	[−0.026, 0.175]	p=0.156

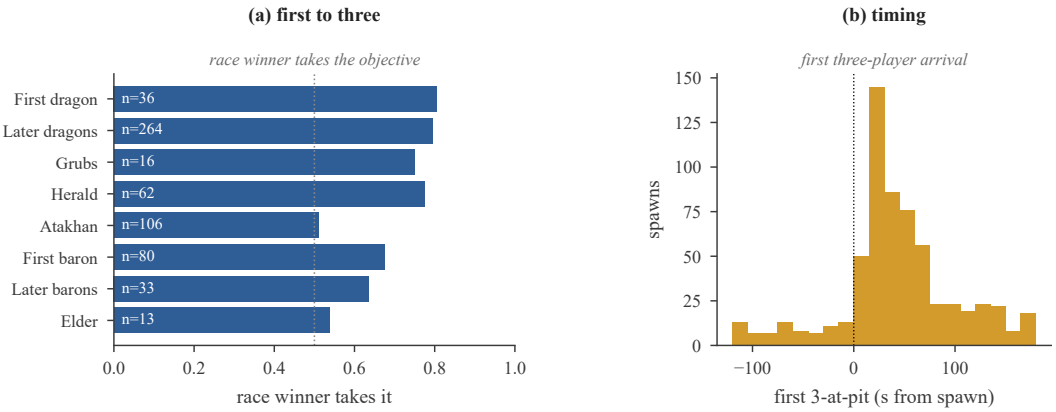


Figure 15: **The first team to reach three players at the pit usually takes river objectives, but not Atakhan.** (a) Share of spawns with a decided race in which the race winner took the objective within 180 s. (b) Time of the first three-player arrival relative to the spawn.

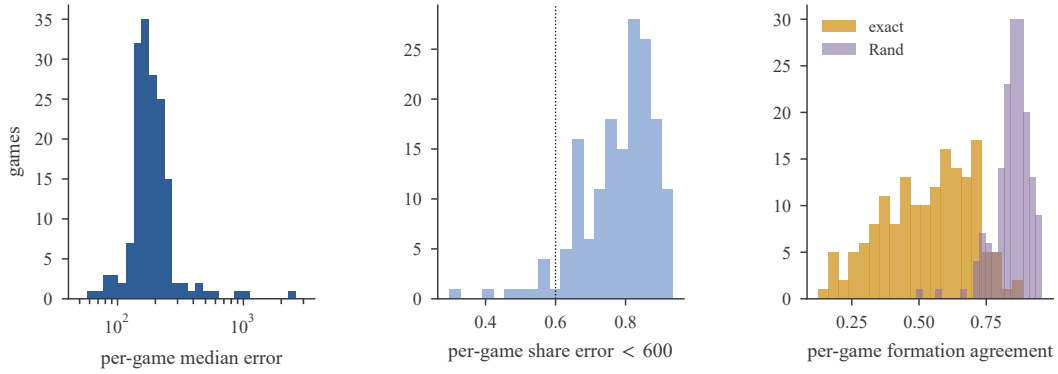


Figure 16: **Most games track well; a few are outliers.** Tracker quality per game. Per-game median error, share of errors below 600 units, and formation agreement with v5 at held-out minutes.

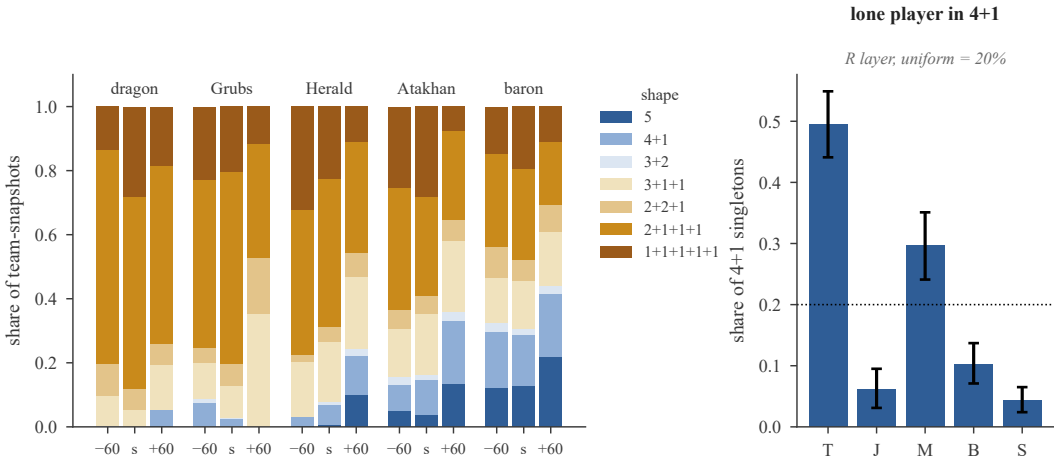


Figure 17: **The lone player in a 4+1 formation is usually the top laner.** Formation shapes on the R layer at $s-60$, s and $s+60$ by objective (left) and the role of the lone player in 4+1 formations against the uniform 20% (right). The top laner is the lone player in 49.5% [44.1, 54.9] of 4+1 formations, the mid laner in 29.7% [24.1, 35.1].

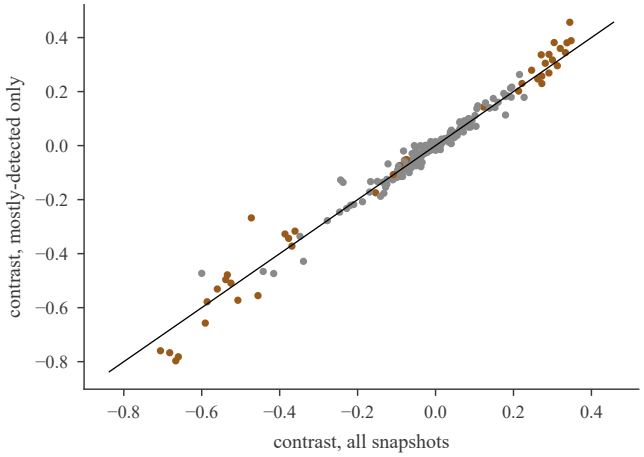


Figure 18: **Dropping mostly-interpolated snapshots leaves the contrasts essentially unchanged.** V-layer trailing-minus-leading contrasts computed on all snapshots and on snapshots with at most half of the positions held or interpolated (radius 1,500; red: headline cells).

Table 13: **Formation vocabularies on the R layer, by objective and time.** Formation vocabulary on the R layer (five alive, risk set). Rarefied = expected distinct partitions at the smallest n.

Objective	Time	n	Partitions	Shapes	Rarefied
Atakhan	$s-60$	261	35	7	24.3
Atakhan	s	283	39	7	23.6
Atakhan	$s+60$	198	36	7	27.0
Baron	$s-60$	239	41	7	27.8
Baron	s	256	37	7	25.3
Baron	$s+60$	200	37	7	27.4
Dragon	$s-60$	304	21	5	12.8
Dragon	s	317	18	4	11.8
Dragon	$s+60$	266	20	6	14.3
Grubs	$s-60$	294	26	6	15.8
Grubs	s	317	22	6	14.1
Grubs	$s+60$	17	9	4	–
Herald	$s-60$	289	24	6	16.8
Herald	s	286	31	7	21.2
Herald	$s+60$	90	23	7	23.0

Table 14: $K = 6$ **is best among** $K = 3, \ldots, 6$ **on the 7-snapshot grid and at both radii.** Discovery-layer robustness: held-out log-likelihood per setup for $K = 3..6$ and the K chosen by each criterion.

Layer	K=3	4	5	6	best held-out	BIC	ICL
main (13 snapshots, radius 1,900)	-83.83	-82.02	-80.82	-80.00	6	6	6
7 snapshots $s-90..s$	-50.89	-49.89	-49.20	-48.70	6	6	6
formation radius 1,500	-86.76	-85.13	-83.27	-82.11	6	6	6
formation radius 2,500	-78.85	-77.00	-76.01	-75.51	6	6	6

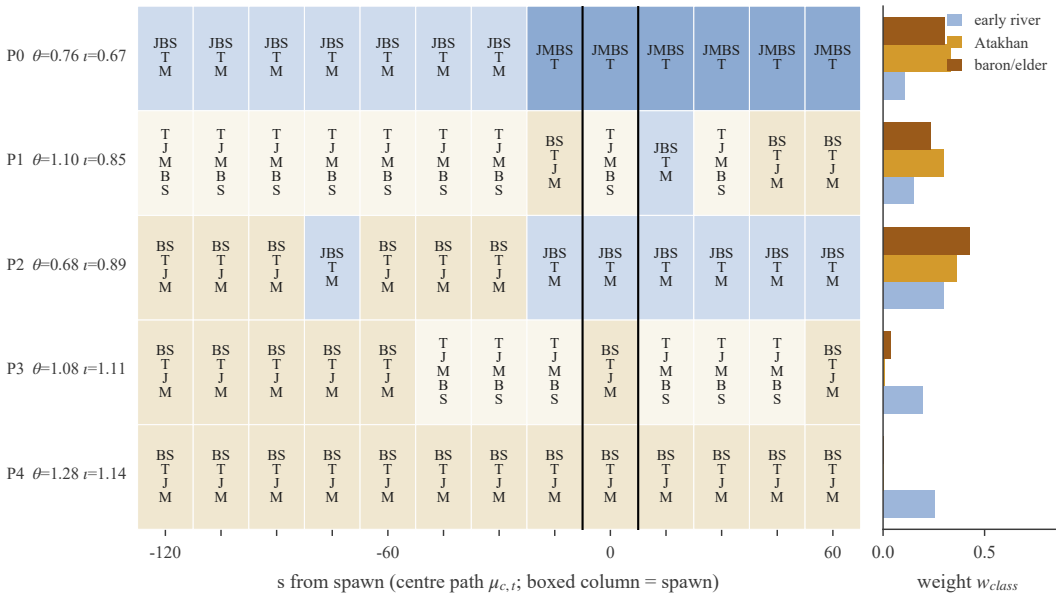


Figure 19: **The** $K = 5$ **dictionary keeps the early-river and late patterns.** Pattern dictionary at $K = 5$ ($\xi = 0.127$).

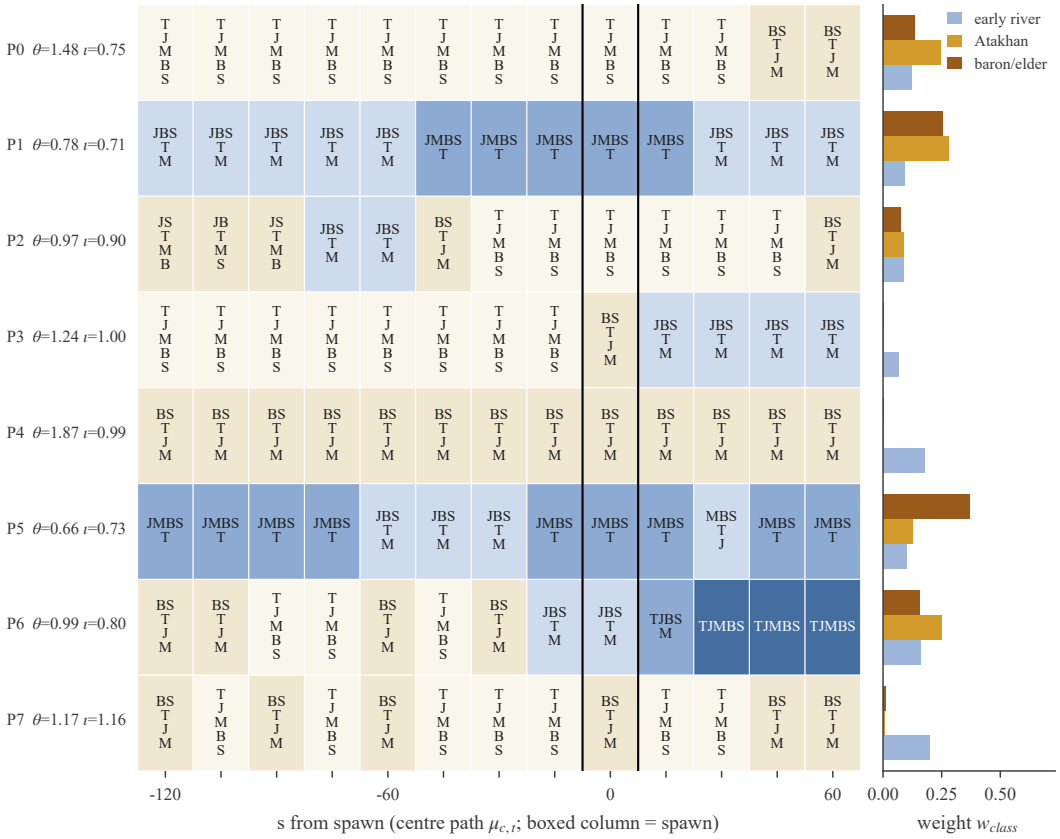


Figure 20: **The $K = 8$ dictionary keeps the early-river and late patterns.** Pattern dictionary at $K = 8$ ($\xi = 0.221$).

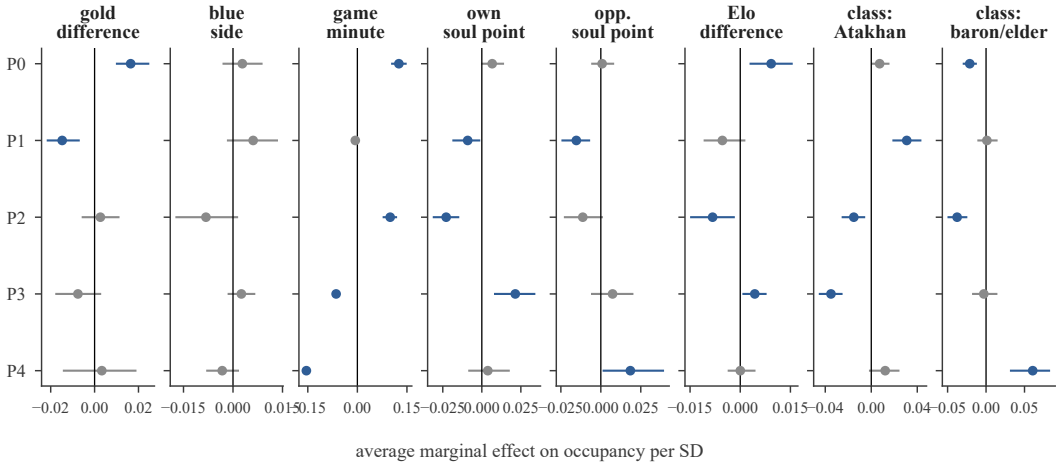


Figure 21: **Context effects are reported at $K = 5$ as a robustness check.** Context effects at $K = 5$.

Table 15: **Context effects on pattern occupancy are mostly small.** Soft-label multinomial logit of pattern occupancy (reference pattern 5), coefficients per SD; * = 95% interval excludes 0 (500 series-block resamples).

	gold k	blue	minute	own soul pt	opp soul pt	elo diff	cls1	cls2
P0	-0.19*	-0.00	-0.80*	-0.03	-0.06	-0.07*	+0.13*	+0.18*
P1	-0.12	-0.05	-4.65*	+0.10	+0.25*	-0.01	+0.10	+1.11*
P2	-0.16*	-0.05	-0.51*	-0.09*	-0.04	-0.08*	-0.06	-0.10*
P3	-0.11*	-0.05*	+0.22*	-0.10*	-0.06	-0.08*	-0.16*	+0.09
P4	-0.36*	+0.01	-3.59*	+0.42*	+0.32*	+0.03	-0.59*	+0.23

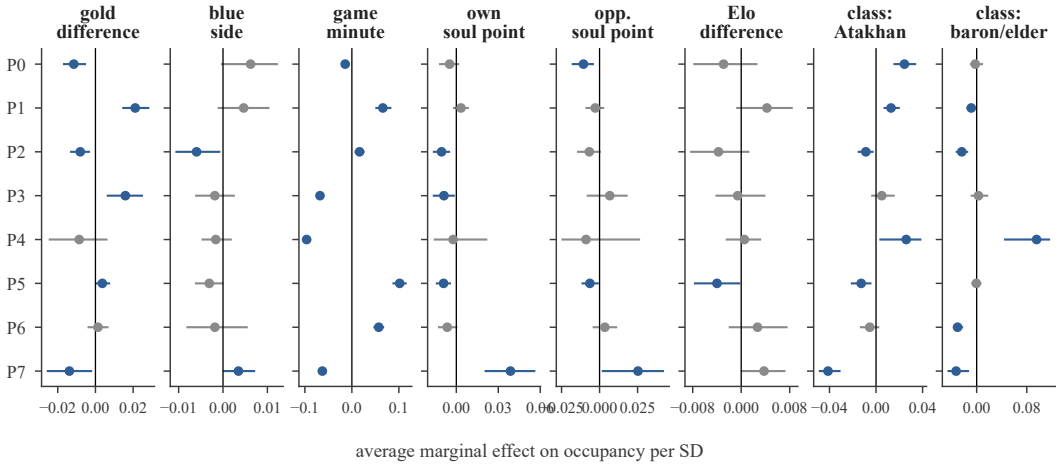


Figure 22: **Context effects are reported at $K = 8$ as a robustness check.** Context effects at $K = 8$.

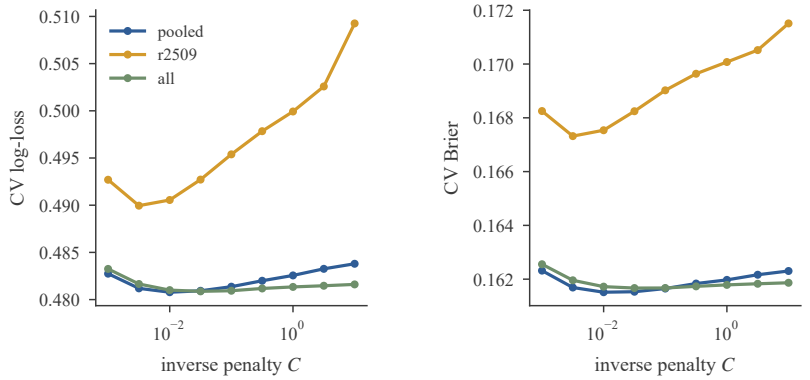


Figure 23: **Cross-validated loss is flat near the chosen penalty.** Cross-validated log-loss and Brier score of W against the inverse penalty C for the three specifications.

Table 16: **The win surface fits out of fold in every game phase.** Out-of-fold fit of the outcome-only win surface W by game phase.

W specification	LL 1–13	14–24	25+	Brier 1–13	14–24	25+	max cal. dev. (pp)
pooled, rule-specific objective terms (main)	0.625	0.439	0.317	0.218	0.143	0.100	1.3
same-rule games only	0.657	0.455	0.288	0.233	0.150	0.093	3.7
all games, shared terms	0.624	0.439	0.319	0.218	0.143	0.101	1.3

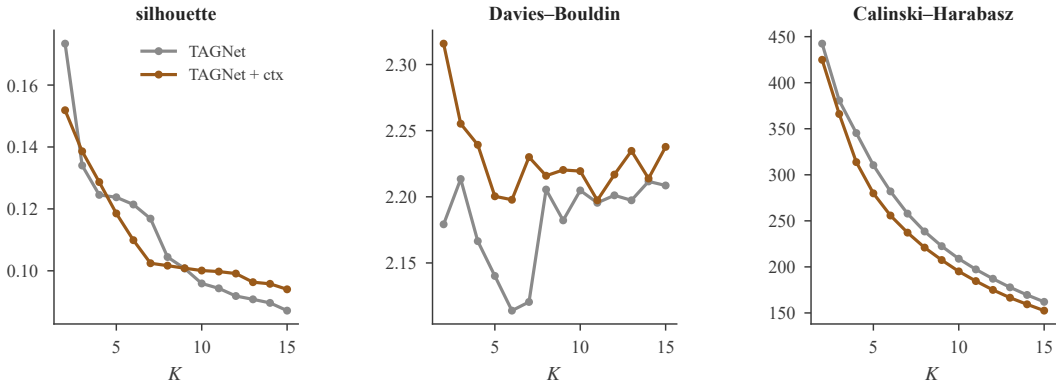


Figure 24: **TAGNet's own criteria fall as K grows.** TAGNet's own cluster criteria on our setups, $K = 2..8$, with and without the context vector.

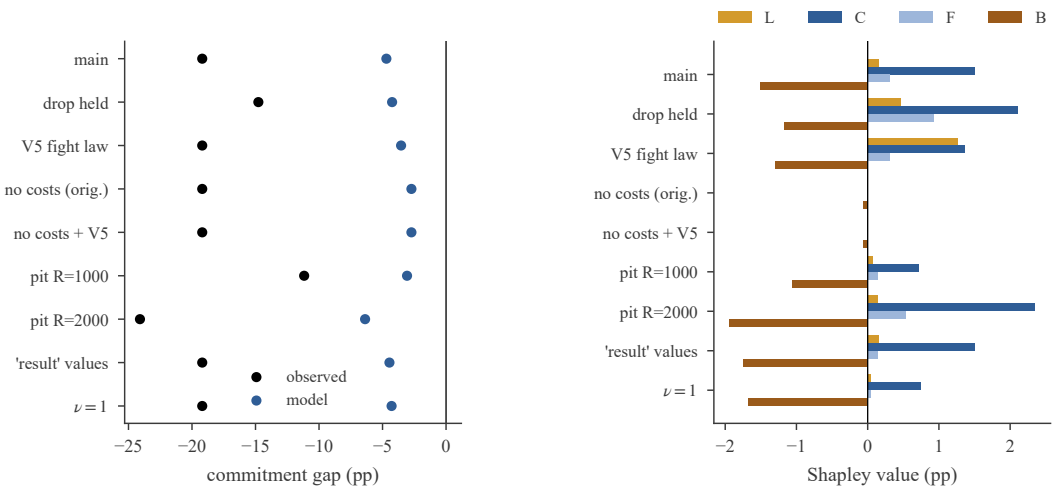


Figure 25: **Gaps and Shapley signs hold across the six non-degenerate specification variants.** LB-QRE specification variants: observed and model commitment gaps (left) and the four Shapley values (right).

Table 17: **All Shapley signs agree with the main fit in the six non-degenerate variants.** LB-QRE specification variants. All Shapley signs agree with the main fit in the six non-degenerate variants; the two no-cost variants collapse to $\lambda = 0$.

Variant	n	ℓ	λ	ν	Obs. gap	Model gap	ϕ_L	ϕ_C	ϕ_F	ϕ_B
main	1,454	-8,343	4.30	0.018	−0.192	−0.047	0.0015	0.0150	0.0030	−0.0151
drop held/interp.	762	-4,264	12.06	0.017	−0.148	−0.042	0.0047	0.0211	0.0092	−0.0117
V5 fight law	1,454	-8,318	4.78	0.121	−0.192	−0.035	0.0126	0.0137	0.0032	−0.0130
no costs (original spec.)	1,454	-9,254	0.00	0.993	−0.192	−0.027	0.0000	0.0000	0.0000	−0.0006
no costs + V5 law	1,454	-9,254	0.00	1.006	−0.192	−0.027	0.0000	0.0000	0.0000	−0.0006
pit radius 1,000	1,451	-7,732	3.14	0.027	−0.112	−0.031	0.0008	0.0072	0.0014	−0.0106
pit radius 2,000	1,457	-8,822	5.52	0.017	−0.241	−0.064	0.0015	0.0235	0.0054	−0.0194
'result' values	1,454	-8,386	2.91	0.024	−0.192	−0.045	0.0016	0.0151	0.0014	−0.0175
$\nu = 1$ (H_{winmax})	1,454	-8,403	0.73	1.000	−0.192	−0.043	0.0004	0.0075	0.0005	−0.0168

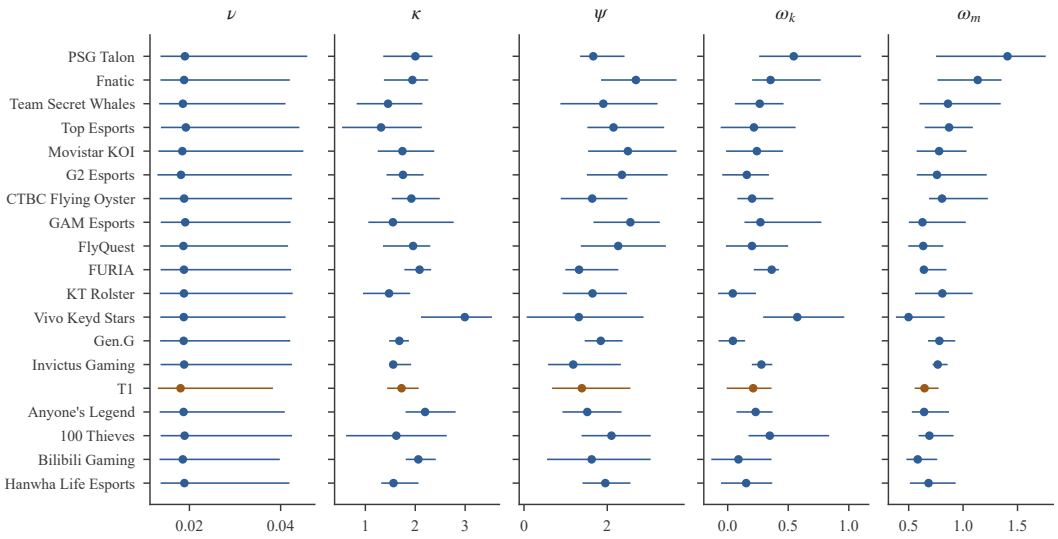


Figure 26: **Team parameters overlap widely.** Team cost and valuation parameters with 95% intervals (T1 in red), teams ordered by λ.

Table 18: **Team parameters come with wide intervals.** Team LB-QRE parameters with 95% series-block bootstrap intervals (shrunk towards the global values).

Team	Contests	λ	κ	ψ	ω_k	ω_m
T1	421	3.3 [1.8, 6.3]	1.73 [1.44, 2.07]	1.39 [0.67, 2.55]	0.21 [-0.01, 0.36]	0.65 [0.56, 0.78]
Gen.G	288	3.7 [2.2, 7.3]	1.68 [1.48, 1.87]	1.84 [1.46, 2.37]	0.04 [-0.08, 0.14]	0.78 [0.68, 0.93]
Anyone's Legend	281	2.8 [1.0, 7.8]	2.20 [1.81, 2.81]	1.52 [0.92, 2.34]	0.23 [0.07, 0.37]	0.64 [0.53, 0.87]
G2 Esports	265	6.2 [3.2, 12.8]	1.75 [1.42, 2.17]	2.35 [1.51, 3.45]	0.16 [-0.04, 0.34]	0.76 [0.57, 1.22]
Bilibili Gaming	252	2.6 [1.3, 6.5]	2.06 [1.81, 2.41]	1.63 [0.55, 3.04]	0.09 [-0.14, 0.36]	0.58 [0.48, 0.76]
CTBC Flying Oyster	220	6.0 [2.8, 12.9]	1.92 [1.53, 2.49]	1.64 [0.88, 2.49]	0.20 [0.08, 0.38]	0.81 [0.69, 1.23]
FlyQuest	177	4.5 [1.9, 9.0]	1.96 [1.35, 2.30]	2.26 [1.36, 3.41]	0.20 [-0.01, 0.50]	0.63 [0.50, 0.82]
Movistar KOI	164	9.9 [3.6, 16.8]	1.74 [1.25, 2.38]	2.50 [1.54, 3.66]	0.24 [-0.01, 0.46]	0.78 [0.57, 1.03]
KT Rolster	150	4.0 [2.9, 7.5]	1.47 [0.95, 1.90]	1.65 [0.93, 2.47]	0.04 [-0.08, 0.23]	0.81 [0.56, 1.09]
Top Esports	124	10.3 [6.0, 16.3]	1.31 [0.53, 2.13]	2.15 [1.52, 3.37]	0.22 [-0.06, 0.56]	0.87 [0.65, 1.09]
GAM Esports	115	5.1 [2.3, 13.3]	1.55 [1.06, 2.77]	2.56 [1.67, 3.26]	0.27 [0.14, 0.77]	0.63 [0.50, 1.02]
FURIA	92	4.3 [3.8, 6.7]	2.09 [1.78, 2.32]	1.32 [0.99, 2.27]	0.36 [0.22, 0.42]	0.64 [0.61, 0.85]
Hanwha Life Esports	91	2.3 [1.9, 4.8]	1.56 [1.32, 2.06]	1.95 [1.41, 2.56]	0.15 [-0.05, 0.37]	0.68 [0.51, 0.93]
Vivo Keyd Stars	59	3.9 [1.9, 11.2]	2.99 [2.11, 3.54]	1.32 [0.06, 2.87]	0.57 [0.29, 0.96]	0.50 [0.38, 0.83]
Team Secret Whales	53	11.8 [4.3, 21.7]	1.45 [0.82, 2.14]	1.90 [0.87, 3.21]	0.26 [0.06, 0.46]	0.86 [0.60, 1.35]
100 Thieves	44	2.7 [1.8, 20.8]	1.62 [0.61, 2.63]	2.10 [1.38, 3.04]	0.35 [0.17, 0.84]	0.69 [0.59, 0.91]
PSG Talon	40	23.9 [4.6, 46.1]	2.00 [1.36, 2.35]	1.66 [1.35, 2.42]	0.55 [0.26, 1.10]	1.41 [0.75, 1.76]
Fnatic	36	14.3 [3.3, 28.3]	1.94 [1.37, 2.26]	2.69 [1.85, 3.66]	0.35 [0.20, 0.77]	1.13 [0.76, 1.35]
Invictus Gaming	36	3.4 [2.4, 7.0]	1.56 [1.50, 1.92]	1.18 [0.58, 2.33]	0.28 [0.20, 0.37]	0.77 [0.72, 0.86]

Table 19: **Split-half reliability of team pattern usage is low.** Pattern usage by team ($K = 6$). Intervals in Fig. 12; split-half reliability is low.

Team	Setups	P0	P1	P2	P3	P4	P5	TV to league
T1	394	0.21	0.20	0.16	0.13	0.09	0.21	0.065
Gen.G	266	0.20	0.18	0.18	0.13	0.09	0.21	0.038
Anyone's Legend	260	0.24	0.15	0.21	0.14	0.08	0.18	0.043
G2 Esports	258	0.18	0.18	0.18	0.14	0.11	0.22	0.062
Bilibili Gaming	244	0.18	0.17	0.20	0.15	0.09	0.22	0.042
CTBC Flying Oyster	208	0.24	0.16	0.17	0.14	0.10	0.19	0.056
FlyQuest	165	0.22	0.16	0.19	0.16	0.10	0.18	0.042
Movistar KOI	161	0.19	0.15	0.24	0.14	0.07	0.21	0.048
KT Rolster	147	0.19	0.21	0.15	0.12	0.09	0.24	0.099
Top Esports	121	0.22	0.19	0.19	0.12	0.10	0.19	0.051
GAM Esports	113	0.17	0.19	0.18	0.18	0.09	0.19	0.072
FURIA	88	0.20	0.17	0.22	0.17	0.06	0.17	0.048
Hanwha Life Esports	82	0.20	0.13	0.20	0.18	0.06	0.22	0.070
Vivo Keyd Stars	55	0.17	0.17	0.24	0.19	0.07	0.16	0.090
Team Secret Whales	51	0.24	0.17	0.24	0.14	0.05	0.16	0.076
100 Thieves	40	0.16	0.18	0.23	0.13	0.08	0.22	0.063
PSG Talon	38	0.23	0.15	0.26	0.14	0.09	0.13	0.092
Fnatic	36	0.32	0.10	0.28	0.10	0.05	0.16	0.200
Invictus Gaming	31	0.16	0.20	0.15	0.14	0.10	0.24	0.104

Table 20: **Faker arrives earlier but is no more central in the formation.** A3: Faker against other starting mid laners in pre-spawn snapshots ($n = 432$ vs 2,545).

Measure	Faker	Other mids	Diff.	95% CI
in pit block at s	0.039	0.031	0.008	[−0.000, 0.018]
arrival rank	2.877	2.993	−0.116	[−0.226, −0.027]
geodesic distance to pit	5,563	6,033	−469	[−709, −275]
alone	0.631	0.622	0.009	[−0.039, 0.063]
in main block	0.337	0.333	0.004	[−0.042, 0.049]
MST degree	1.481	1.474	0.007	[−0.023, 0.041]
MST betweenness	0.259	0.253	0.006	[−0.011, 0.023]

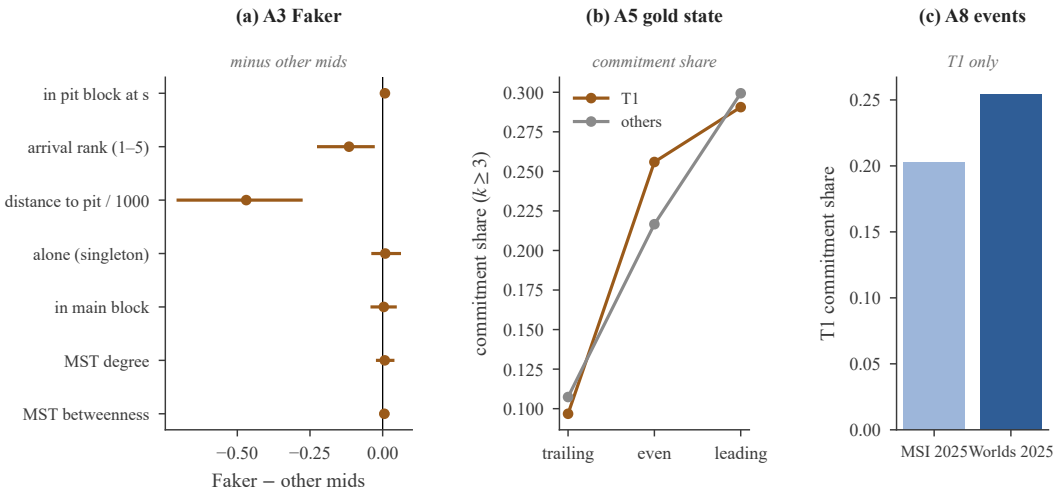


Figure 27: **Faker arrives earlier, but T1 behaves like the rest of the league.** T1 and Faker in numbers. (a) A3: Faker minus other starting mid laners on seven position measures in pre-spawn snapshots (distance in thousands of units). (b) A5: commitment share by gold state, T1 against all other teams. (c) A8: T1's commitment share at MSI and Worlds 2025.

Table 21: **T1's behaviour by gold state matches the rest of the league.** A5: T1 and all other teams by gold state at the commitment time (commit = $k \geq 3$; polarisation = $k \in \{0, 5\}$).

Group	State	n	Commit	Polar.	Mean k
T1	trailing	93	0.097	0.731	0.60
others	trailing	652	0.107	0.724	0.60
T1	even	211	0.256	0.460	1.31
others	even	1205	0.217	0.501	1.14
T1	leading	117	0.291	0.274	1.57
others	leading	628	0.299	0.277	1.64

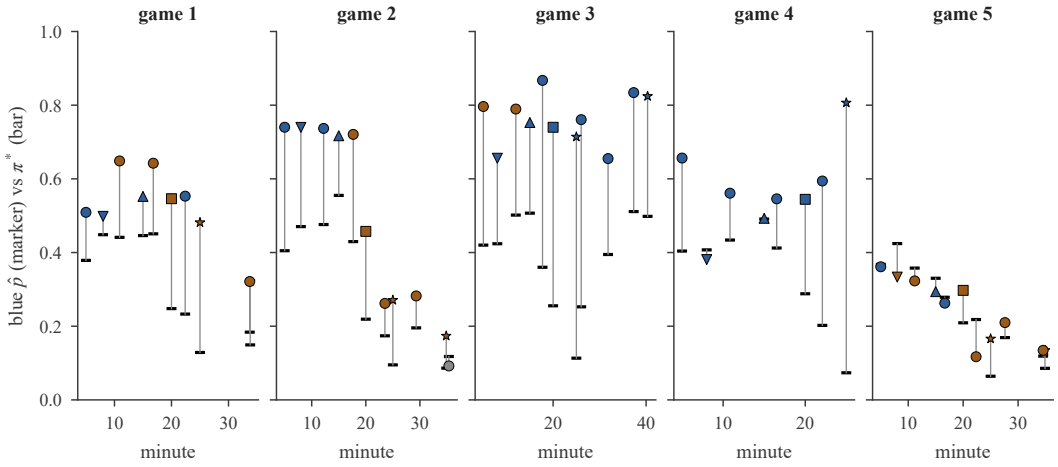


Figure 28: **The ledger lists every spawn of the Worlds 2025 final.** A7: every spawn of the Worlds 2025 final. Markers: blue's fight probability $\hat{p}$ (colour = taker: blue/red/none; shape = objective); bars end at the break-even π^*.

Table 22: **The ledger lists every spawn of the Worlds 2025 final.** A7: decision ledger of the Worlds 2025 final (T1 3–2 KT Rolster). Actions (k, m, τ) at the commitment time, blue's fight probability $\hat{p}$, break-even π^* and the objective's value to blue (pts).

G	Spawn	Min	W_B	GD$_B$	Blue	Red	$\hat{p}$	π^*	V_B	Took
1	Drag 1	5.0	0.54	+21	(2,1,0)	(0,1,0)	0.51	0.38	+5.8	B
1	Grubs 1	8.0	0.66	-7	(1,1,1)	(0,3,0)	0.50	0.45	+3.5	B
1	Drag 2	10.9	0.74	+302	(3,0,0)	(2,0,0)	0.65	0.44	+6.0	R
1	Herald 1	15.0	0.64	+71	(3,0,0)	(0,0,0)	0.55	0.45	+5.5	B
1	Drag 3	16.8	0.82	+1,472	(0,2,0)	(3,1,0)	0.64	0.45	+5.7	R
1	Atak 1	20.0	0.63	+564	(2,0,0)	(0,0,0)	0.55	0.25	+17.6	R
1	Drag 4	22.4	0.63	+713	(2,0,0)	(3,0,0)	0.55	0.23	+9.6	B
1	Baron 1	25.0	0.45	-278	(0,0,0)	(3,0,0)	0.48	0.13	+33.9	R
1	Drag 6	33.7	0.19	-4,059	(0,0,0)	(1,1,1)	0.32	0.18	+14.6	R
1	Baron 2	33.8	0.19	-4,059	(0,0,0)	(3,0,0)	0.32	0.15	+39.2	–
2	Drag 1	5.0	0.65	+616	(0,1,0)	(0,1,0)	0.74	0.40	+5.2	B
2	Grubs 1	8.0	0.73	+1,075	(1,2,0)	(0,4,0)	0.74	0.47	+3.0	B
2	Drag 2	12.2	0.80	+1,722	(1,1,0)	(0,1,0)	0.74	0.48	+5.2	B
2	Herald 1	15.0	0.87	+1,976	(3,1,0)	(0,2,0)	0.72	0.56	+2.5	B
2	Drag 3	17.7	0.90	+2,485	(3,0,0)	(0,0,0)	0.72	0.43	+7.8	R
2	Atak 1	20.0	0.42	-509	(4,0,0)	(1,1,1)	0.46	0.22	+21.3	R
2	Drag 4	23.5	0.14	-3,860	(3,0,0)	(3,1,0)	0.26	0.17	+15.6	R
2	Baron 1	25.0	0.16	-3,823	(0,1,0)	(3,1,0)	0.27	0.10	+35.9	R
2	Drag 5	29.3	0.16	-4,272	(3,0,1)	(0,0,0)	0.28	0.20	+15.0	R
2	Baron 2	34.9	0.03	-8,952	(0,0,0)	(3,0,0)	0.17	0.09	+23.0	R
2	Drag 6	35.4	0.03	-8,952	(0,1,0)	(0,4,0)	0.09	0.12	+4.6	–
3	Drag 1	5.0	0.70	+804	(0,1,0)	(2,1,0)	0.80	0.42	+4.7	R
3	Grubs 1	8.0	0.57	+629	(1,1,0)	(0,1,0)	0.66	0.42	+3.9	B
3	Drag 2	12.0	0.85	+2,131	(0,0,0)	(1,1,0)	0.79	0.50	+3.6	R
3	Herald 1	15.0	0.79	+2,335	(1,0,1)	(0,0,0)	0.75	0.51	+3.9	B
3	Drag 3	17.7	0.87	+3,502	(1,0,0)	(0,0,0)	0.87	0.36	+4.0	B
3	Atak 1	20.0	0.67	+2,255	(2,0,0)	(0,0,0)	0.74	0.26	+16.1	B
3	Baron 1	25.0	0.91	+4,347	(3,1,0)	(0,5,0)	0.71	0.11	+12.3	B
3	Drag 4	26.0	0.92	+4,807	(3,0,0)	(0,2,0)	0.76	0.25	+3.1	B
3	Drag 5	31.8	0.85	+3,332	(3,1,0)	(0,4,0)	0.66	0.39	+8.7	B
3	Drag 6	37.3	0.98	+9,905	(3,0,0)	(1,0,0)	0.83	0.51	+9.1	B
3	Baron 2	40.3	0.97	+9,097	(4,0,0)	(0,0,0)	0.82	0.50	+4.7	B
4	Drag 1	5.0	0.64	+372	(0,1,0)	(2,1,0)	0.66	0.40	+5.2	B
4	Grubs 1	8.0	0.50	-482	(1,1,1)	(0,1,0)	0.38	0.41	+4.0	B
4	Drag 2	10.8	0.73	+353	(2,2,0)	(1,1,0)	0.56	0.43	+6.2	B
4	Herald 1	15.0	0.75	-58	(2,1,0)	(0,3,0)	0.49	0.49	+4.4	B
4	Drag 3	16.5	0.83	+422	(2,1,0)	(4,0,0)	0.55	0.41	+11.4	B
4	Atak 1	20.0	0.83	+504	(4,0,0)	(4,0,0)	0.54	0.29	+9.6	B
4	Drag 4	22.1	0.92	+1,196	(4,0,0)	(3,0,0)	0.59	0.20	+17.2	B
4	Baron 1	25.0	0.99	+4,488	(3,1,0)	(2,0,0)	0.81	0.07	+3.0	B
5	Drag 1	5.0	0.49	-320	(4,1,0)	(0,1,0)	0.36	0.37	+6.0	B
5	Grubs 1	8.0	0.57	-712	(1,1,1)	(1,1,0)	0.33	0.42	+3.9	R
5	Drag 2	11.2	0.48	-1,095	(0,1,0)	(2,1,0)	0.32	0.36	+8.6	R
5	Herald 1	15.0	0.28	-1,800	(3,0,0)	(0,1,0)	0.29	0.33	+5.3	B
5	Drag 3	16.7	0.23	-2,446	(3,0,0)	(0,1,0)	0.26	0.28	+8.8	B
5	Atak 1	20.0	0.34	-2,489	(0,0,0)	(3,0,0)	0.30	0.21	+21.2	R
5	Drag 4	22.4	0.45	-2,431	(0,0,0)	(3,0,1)	0.12	0.22	+21.2	R
5	Baron 1	25.0	0.04	-6,216	(0,4,0)	(0,5,0)	0.17	0.06	+24.2	R
5	Drag 5	27.6	0.10	-5,733	(1,0,1)	(3,0,0)	0.21	0.17	+11.8	R
5	Drag 6	34.6	0.03	-10,456	(0,0,0)	(1,0,1)	0.13	0.12	+4.7	R
5	Baron 2	34.9	0.03	-10,456	(0,0,0)	(3,0,0)	0.13	0.09	+22.7	–